Sri Lanka Institute of Advanced Technological Education (SLIATE)

Final Year Individual Project



Web-Based Travel & Tour Management System with Real-Time Vehicle Tracking

For the partial fulfilment of individual project in Second year Second semester of Higher National Diploma in Information Technology in Advanced Technological Institute of Tangalle

Submitted By: H.P.D Dilshan Kumara.

Supervisor: Mr. MAM Althaf

Registration No: TAN/IT/2324/F/0062

ABSTRACT

The **Travel & Tour Management System** is a complete web-based software that helps a travel company manage all its daily work in one place. The system handles tour packages, customer bookings, vehicle fleet management, live GPS tracking, customer emails, and business reports.

In normal travel agencies, most work is done by hand — writing bookings on paper, calling customers one by one, and checking vehicles by phone. This causes many problems like double bookings, missed emails, and no clear view of what is happening. Our new system solves all these problems.

The system is built using **PHP** (backend), **HTML5, CSS3, JavaScript** (frontend), and **MySQL** (database). It also uses **PHPMailer** for sending emails, **Dompdf** for creating PDF reports, **Chart.js** for charts, **DataTables** for tables, **Google Maps API** for route maps, and **Leaflet.js** for live vehicle tracking.

The system has **three different user roles**:

1. **Administrator** — Full control of the whole business
2. **Vehicle Manager** — Controls only the fleet of vehicles
3. **Customer** — Books tours and tracks their trips

The most special feature of this system is the **Real-Time Trip Status Engine**. This engine automatically decides if a booking is **Upcoming**, **Ongoing**, or **Completed** by looking at today's date, the booking date, and the number of travel days. No human needs to change the status manually.

By using this system, a travel agency can save time, reduce mistakes, send automatic emails, track vehicles live on a map, and get clear business reports with just one click.

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CHAPTER 01: INTRODUCTION

1.1 Organizational Background

The travel and tourism industry is growing fast all around the world. In Sri Lanka, there are many travel agencies, tour operators, and vehicle rental companies. Most of these are small or medium businesses.

Today, most small travel agencies still use **old methods**:

- Writing bookings in paper registers
- Calling customers by phone to confirm bookings
- Sending drivers by phone to check vehicle status
- Writing reports by hand in notebooks

These old methods cause many problems — mistakes, slow service, unhappy customers, and no clear understanding of the business.

Travel & Tour is a modern travel company that offers:

- Tour packages (pre-made tours)
- Custom trip planning (customer designs their own trip)
- Vehicle rental services
- Customer support
- Payment handling

The company wanted to **change from manual work to digital work**. So they built the **Travel & Tour Management System** — a web application that does everything in one place.

1.2 Project Background

Before this project, the company faced these problems:

Problem 1 — Double Bookings: Sometimes two customers booked the same vehicle for the same date because there was no way to check.

Problem 2 — Slow Confirmation: After a customer booked, someone had to call or email them manually. Sometimes this took hours or days.

Problem 3 — No Real-Time View: The admin didn't know if a trip was ongoing or finished until the driver called to say so.

Problem 4 — Manual Reports: To see how much money came in this month, someone had to count everything by hand.

Problem 5 — Lost Data: Old paper records were easily lost or damaged.

To solve all these problems, we built this new system with the following features:

- Online tour package management
- Automatic booking confirmation emails
- Real-time trip status tracking
- Live GPS tracking of all vehicles
- Automatic PDF reports
- Full fleet management with Kanban boards
- Custom trip planner with Google Maps
- Advanced analytics dashboard

1.3 Aims and Objectives

1.3.1 Aims

To design, develop, and implement a web-based Travel & Tour Management System that:

- Automates all daily travel agency operations
- Provides real-time tracking of bookings and vehicles
- Sends automatic email notifications
- Generates detailed business reports
- Offers three different user roles with specific features

1.3.2 Objectives

Here are the main objectives of this project:

1. **Replace paper registers** with a proper MySQL database system
2. **Build a modern login screen** with video background and animated design
3. **Create secure login** for three roles: Admin, Vehicle Manager, Customer
4. **Enable real-time trip status** calculation (Upcoming/Ongoing/Completed)
5. **Automate emails** for booking confirmations and feedback replies
6. **Create PDF reports** for business analytics
7. **Build live GPS tracking** with Leaflet.js map
8. **Implement GPS simulation** for testing and demonstration
9. **Create advanced reporting suite** with nine tabs
10. **Provide Kanban board** for visual booking management
11. **Enable custom trip planning** with Google Maps integration
12. **Support multiple payment methods** with confetti animation
13. **Track vehicle expenses** and maintenance schedules
14. **Maintain audit logs** for all important changes

1.4 Problem Statement

The main problem this system solves is: *"How can a small travel agency automate all its daily operations without needing expensive enterprise software?"*

Traditional software solutions cost a lot of money. They are complicated and not customizable. This system solves this by offering a complete solution built with **free, open-source technologies**.

1.5 Scope of the Project

What is included:

- Admin panel with 16 files
- User panel with 9 files
- Vehicle manager panel with 12 files
- Live GPS tracking system
- Email notification system
- PDF report generation
- Advanced reporting with charts

What is NOT included:

- Real credit card payment gateway (simulated only)
- Mobile app (web only)
- Multi-language support
- Cloud deployment (local server only)

CHAPTER 02: LITERATURE REVIEW

2.1 Introduction

In the last 10 years, the tourism industry has changed a lot. Most work has moved from paper to computers. Research shows that web-based management systems help businesses in many ways:

- **Save time** — Automate repetitive tasks
- **Reduce mistakes** — Data is validated before saving
- **Improve customer service** — Customers get instant confirmations
- **Better reporting** — Reports are generated in seconds
- **Track everything** — All actions are logged

2.2 Study of Similar Applications

We studied several existing travel management systems:

1. Static Websites (Basic Level)

- Show tour packages in a list
- No booking capability
- No email automation
- No admin panel

2. Offline POS Software (Medium Level)

- Handles billing inside the office
- No customer-facing features
- No online booking
- No live tracking

3. Enterprise Systems (High Level)

- Full-featured (like Booking.com, Expedia)
- Very expensive
- Not customizable
- Too complicated for small agencies

2.3 Comparison with Existing Systems

Feature	Static Website	Offline POS	Enterprise	Our System
Package browsing	Yes	No	Yes	Yes
Online booking	No	No	Yes	Yes
Auto email	No	No	Yes	Yes
Real-time status	No	No	Yes	Yes
Live GPS tracking	No	No	Yes	Yes
Custom trip planner	No	No	Yes	Yes
Kanban board	No	No	No	Yes
PDF reports	No	Yes	Yes	Yes
Multiple user roles	No	No	Yes	Yes
Free / Open source	Yes	No	No	Yes

Our system combines all the best features at zero cost.

CHAPTER 03: METHODOLOGY

3.1 SDLC Phases Applied in the Project

We followed the **Software Development Life Cycle (SDLC)** to build this project. Here are the five phases:

Phase 1: Requirement Analysis & Planning

- Met with travel agency owners to understand their needs
- Listed all required features for each role
- Identified problems in current manual system
- Created a plan for development

Phase 2: System & Database Design

- Designed 21 database tables
- Created wireframes for all screens
- Planned the database relationships (foreign keys)
- Designed the API endpoints

Phase 3: Frontend & Backend Development

- Built 36 PHP files with HTML, CSS, and JavaScript
- Created a modern glassmorphism UI theme
- Integrated Google Maps API and Leaflet.js
- Added Chart.js for data visualization
- Connected PHPMailer for emails
- Added Dompdf for PDF reports

Phase 4: System Testing

- **Unit testing** — Tested each function separately
- **Integration testing** — Tested how modules work together
- **User acceptance testing** — Tested with real users
- **Security testing** — Tested SQL injection and XSS attacks

Phase 5: Deployment & Maintenance

- Deployed on XAMPP local server
- Created user documentation

- Set up backup procedures
- Planned future updates

3.2 Techniques and Materials

3.2.1 Hardware Requirements

Component	Minimum	Recommended
Processor	Intel Core i3	Intel Core i5 or higher
RAM	4 GB	8 GB
Storage	10 GB free space	20 GB free space
Display	1366 × 768	1920 × 1080
Internet	1 Mbps	5 Mbps or higher

3.2.2 Software Requirements

Category	Software

Operating System	Windows 10 / Windows 11
Web Server	Apache (XAMPP)
Backend Language	PHP 7.4+
Frontend	HTML5, CSS3, JavaScript ES6+
Database	MySQL 8.0
Database Tool	phpMyAdmin
Code Editor	Visual Studio Code
Email Library	PHPMailer v7.1
PDF Library	Dompdf v3.1
Charts	Chart.js 4.x
Tables	DataTables 1.13
Maps	Google Maps API + Leaflet.js
Icons	Font Awesome 6.5
Fonts	Google Fonts (Inter, Poppins, Orbitron)

3.2.3 Client PC / Web Requirements

- **Browser:** Chrome, Edge, Firefox, Safari
- **JavaScript:** Must be enabled
- **Cookies:** Must be enabled for session
- **Internet:** Required for maps and email

3.3 Interface Design

3.3.1 Interface Design

Interface design is about making the system look nice and easy to use. We used a **Glassmorphism** design theme which means:

- **Frosted glass effect** — Semi-transparent cards
- **Video background** — Moving video behind everything
- **Animated blobs** — Floating colored circles
- **Floating particles** — Small dots that move
- **Smooth animations** — Everything fades in nicely

Each user role has its own color-coded interface:

- **Admin** — Blue-green gradient theme
- **Vehicle Manager** — Purple-blue theme
- **Customer** — Sky blue-green theme

3.3.2 Computer Design

The system uses a **3-tier architecture**:

Tier 1 — Presentation Layer (Frontend)

- HTML pages with embedded CSS and JavaScript
- Responsive design (works on all screen sizes)
- Single-page application feel with AJAX

Tier 2 — Application Layer (Backend)

- PHP scripts that process requests
- Business logic for all operations
- Session management for login

- Email and PDF generation

Tier 3 — Data Layer (Database)

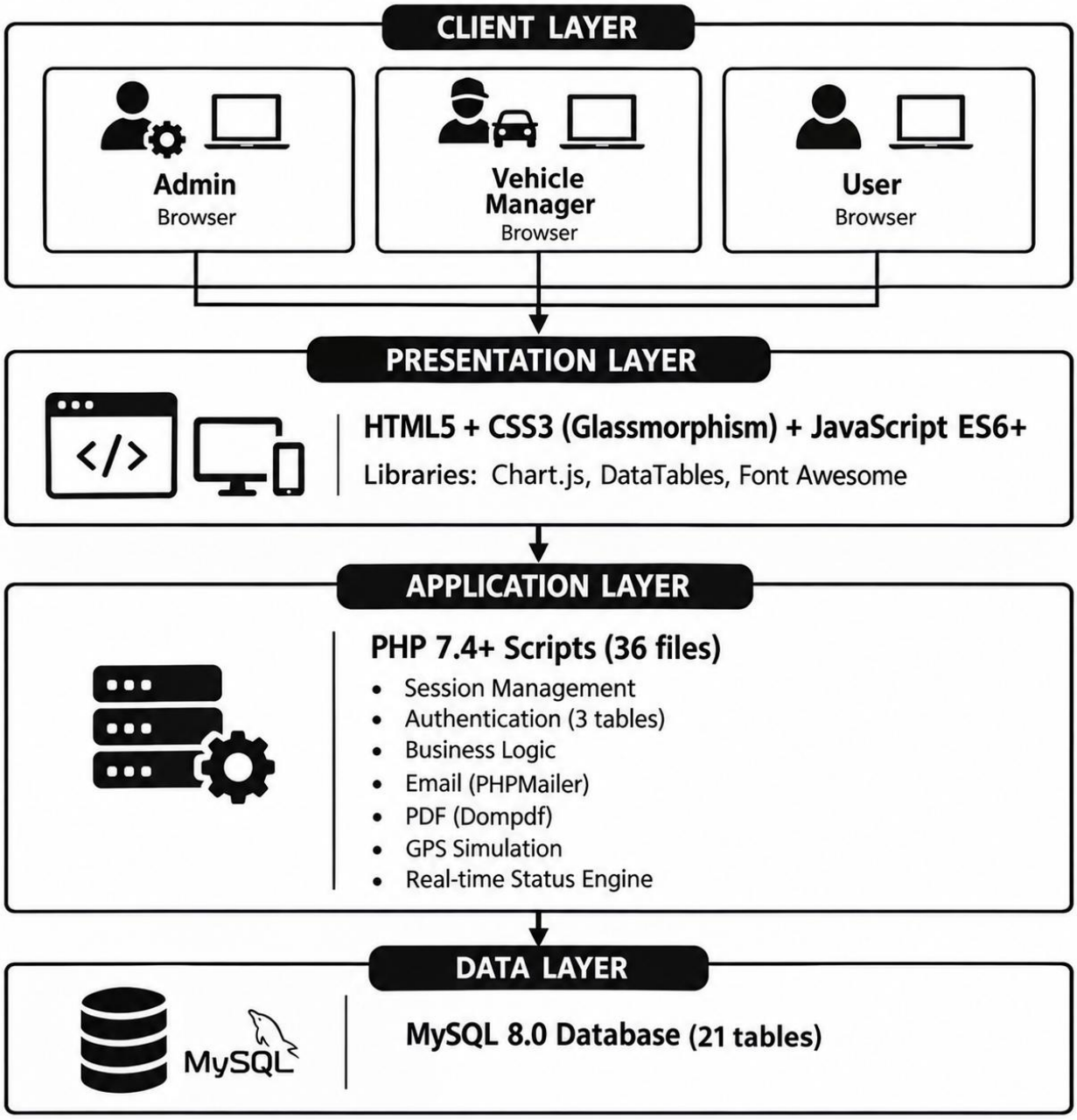
- MySQL database with 21 tables
- Foreign key relationships
- Indexed columns for speed
- Backup and recovery system

3.3.3 Guidelines for Human-Computer Interaction Design

We followed these HCI guidelines:

1. **Consistency** — Same design on every page
2. **Feedback** — Show clear messages for every action
3. **Error Prevention** — Validate inputs before submitting
4. **Recognition over Recall** — Show options, don't make users memorize
5. **Flexibility** — Support keyboard shortcuts
6. **Aesthetic Design** — Modern and beautiful interface
7. **Help Users Recognize Errors** — Clear error messages
8. **Accessibility** — Works on all devices

3.4 System Architecture (3-Tier Detailed)



3.5 Database Design (All 21 Tables Explained)

Table 1: users

Purpose: Stores all customer accounts

Column	Type	Description
id	INT	Primary key, auto increment
full_name	VARCHAR(255)	Customer's full name
email	VARCHAR(191)	Unique email address
phone	VARCHAR(20)	Phone number
address	TEXT	Home address
password	VARCHAR(255)	Bcrypt hashed password
role	ENUM	'user' or 'admin'
created_at	TIMESTAMP	Account creation date

Table 2: admin

Purpose: Stores administrator accounts

Column	Type	Description
AdminID	INT	Primary key
FullName	VARCHAR(100)	Admin's full name
Email	VARCHAR(100)	Unique admin email
Password	VARCHAR(255)	Plain text password
PhoneNumber	VARCHAR(20)	Contact phone
DateCreated	TIMESTAMP	Account creation date

Table 3: vehicle_manager

Purpose: Stores vehicle manager accounts

Column	Type	Description
id	INT	Primary key
name	VARCHAR(100)	Manager's name
email	VARCHAR(100)	Unique email
password	VARCHAR(255)	SHA1 hashed password
phone	VARCHAR(20)	Contact phone
role	ENUM	'admin' or 'manager'
status	ENUM	'active' or 'inactive'
last_login	DATETIME	Last successful login
created_at	TIMESTAMP	Creation date

Table 4: operators

Purpose: Stores vehicle operator companies

Column	Type	Description
id	INT	Primary key
name	VARCHAR(100)	Operator company name
email	VARCHAR(100)	Company email
password	VARCHAR(255)	Bcrypt hashed
phone	VARCHAR(20)	Company phone
status	ENUM	'active' or 'inactive'

Table 5: drivers

Purpose: Stores driver details

Column	Type	Description
id	INT	Primary key
name	VARCHAR(100)	Driver's full name
phone	VARCHAR(20)	Contact phone
license_number	VARCHAR(50)	Driving license number
license_expiry	DATE	License expiry date
address	TEXT	Home address
status	ENUM	available, on_duty, on_leave, inactive
assigned_vehicle_id	INT	Foreign key to vehicles

Table 6: packages

Purpose: Stores all tour packages

Column	Type	Description
id	INT	Primary key
title	VARCHAR(255)	Package name
description	TEXT	Full description
latitude	VARCHAR(50)	Starting point latitude
longitude	VARCHAR(50)	Starting point longitude
latitude2	VARCHAR(50)	End point latitude
longitude2	VARCHAR(50)	End point longitude
price	DECIMAL(10,2)	Price per person
icon	VARCHAR(50)	Emoji icon
status	VARCHAR(20)	Active or Deleted
image	VARCHAR(500)	Package image URL/path

Table 7: bookings

Purpose: Stores all customer bookings

Column	Type	Description
id	INT	Primary key
user_id	INT	Foreign key to users
package_id	INT	Foreign key to packages
package_name	VARCHAR(255)	Package name at booking time
manual_package_name	VARCHAR(255)	Custom package name
booking_date	DATETIME	Date of booking
status	VARCHAR(50)	pending, Approved, Confirmed, Completed, Cancelled, Deleted
pickup_location	VARCHAR(255)	Where customer is picked up
destination_location	VARCHAR(255)	Where customer goes

calculated_distance	DECIMAL(10,2)	Distance in kilometers
total_cost	DECIMAL(10,2)	Total price
number_of_days	INT	Trip duration
vehicle_id	INT	Assigned vehicle
start_date	DATE	Trip start date
end_date	DATE	Trip end date
number_of_members	INT	Number of people
pickup_lat / pickup_lng	DECIMAL	Pickup coordinates
dest_lat / dest_lng	DECIMAL	Destination coordinates
petrol_cost	DECIMAL(12,2)	Petrol cost
vehicle_cost	DECIMAL(12,2)	Vehicle rental cost

Table 8: vehicles

Purpose: Stores all vehicle details

Column	Type	Description
id	INT	Primary key
vehicle_name	VARCHAR(100)	Vehicle model name
category	VARCHAR(50)	Sedan, SUV, Van, Bus
price_per_km	DECIMAL(10,2)	Charge per kilometer
petrol_rate	DECIMAL(10,2)	Petrol charge per liter
capacity	INT	Number of seats
operator_id	INT	Foreign key to operators
status	ENUM	Available, On Trip, Maintenance, Inactive
driver_name	VARCHAR(100)	Driver's name
driver_phone	VARCHAR(20)	Driver's phone
last_lat / last_lng	DECIMAL	Current GPS coordinates
last_update	DATETIME	Last GPS update

next_maintenance	DATE	Next service date
insurance_expiry	DATE	Insurance expiry
registration_expiry	DATE	Registration expiry
fuel_type	VARCHAR(20)	Petrol or Diesel
transmission	VARCHAR(20)	Manual or Auto
driver_id	INT	Foreign key to drivers

Table 9: payments

Purpose: Stores all payment transactions

Column	Type	Description
id	INT	Primary key
user_id	INT	Customer ID
package_id	INT	Package ID
amount	DECIMAL(10,2)	Payment amount
card_name	VARCHAR(255)	Name on card

transaction_id	VARCHAR(50)	Unique transaction ID
status	VARCHAR(20)	Success, Pending, Failed
payment_date	DATETIME	Payment timestamp

Table 10: contact_feedbacks

Purpose: Stores customer feedback and admin replies

Column	Type	Description
id	INT	Primary key
name	VARCHAR(255)	Customer name
email	VARCHAR(255)	Customer email
subject	VARCHAR(255)	Feedback subject
message	TEXT	Feedback message
submitted_at	TIMESTAMP	Submission time
admin_reply	TEXT	Admin's reply

Table 11: admin_notifications

Purpose: Stores notifications for admin

Column	Type	Description
id	INT	Primary key
message	TEXT	Notification text
type	VARCHAR(50)	Notification type
link	VARCHAR(500)	Related page link
is_read	TINYINT	Read status (0 or 1)
created_at	TIMESTAMP	Created time

Table 12: notifications

Purpose: Stores notifications for vehicle managers

Table 13: audit_logs

Purpose: Logs all important admin and manager actions

Table 14: vehicle_audit_logs

Purpose: Detailed logs for vehicle changes

Table 15: system_logs

Purpose: Stores general system events

Table 16: system_settings

Purpose: Stores key-value system settings

Table 17: password_resets

Purpose: Stores password reset tokens

Table 18: messages

Purpose: Stores general contact messages

Table 19: vehicle_documents

Purpose: Stores uploaded vehicle documents

Table 20: vehicle_expenses

Purpose: Stores vehicle expense records

Table 21: booking_vehicles

Purpose: Links bookings with multiple vehicles (for group trips)

3.6 Module Design (Every File Explained)

3.6.1 ADMIN MODULE — 16 Files

File 1: admin/dashboard.php

Purpose: Main control panel showing all key metrics.

What it does:

- Shows 8 KPI cards (Total Users, Travellers, Packages, Active Packages, Bookings, Today's Bookings, Revenue, Average Cost)
- Shows 3 charts (Revenue Trend, Booking Status, Popular Packages)
- Displays real-time vehicle availability alerts
- Shows notification bell with unread count
- Shows quick action buttons

Key Function 1: getValue($conn, $query)

- Purpose: Run a SQL query and get one value
- How it works: Runs query → If fails returns 0 → Otherwise fetches first row → Returns first column value
- Used for: Counting total users, total packages, etc.

Key Function 2: getRow($conn, $query)

- Purpose: Get one complete row from database
- Returns: An associative array with column names as keys
- Used for: Getting top package info, top traveller info

Key Function 3: getRows($conn, $query)

- Purpose: Get multiple rows
- Returns: Array of associative arrays
- Used for: Getting recent bookings, notifications

Auto Vehicle Update Logic:

Every time an admin loads this page, the system:

1. Checks all bookings where vehicle_id IS NOT NULL
2. Looks for bookings where the trip has finished
3. Changes vehicle status from "Booked" to "Available"
4. Creates a notification for the admin
5. Notification appears in the bell icon

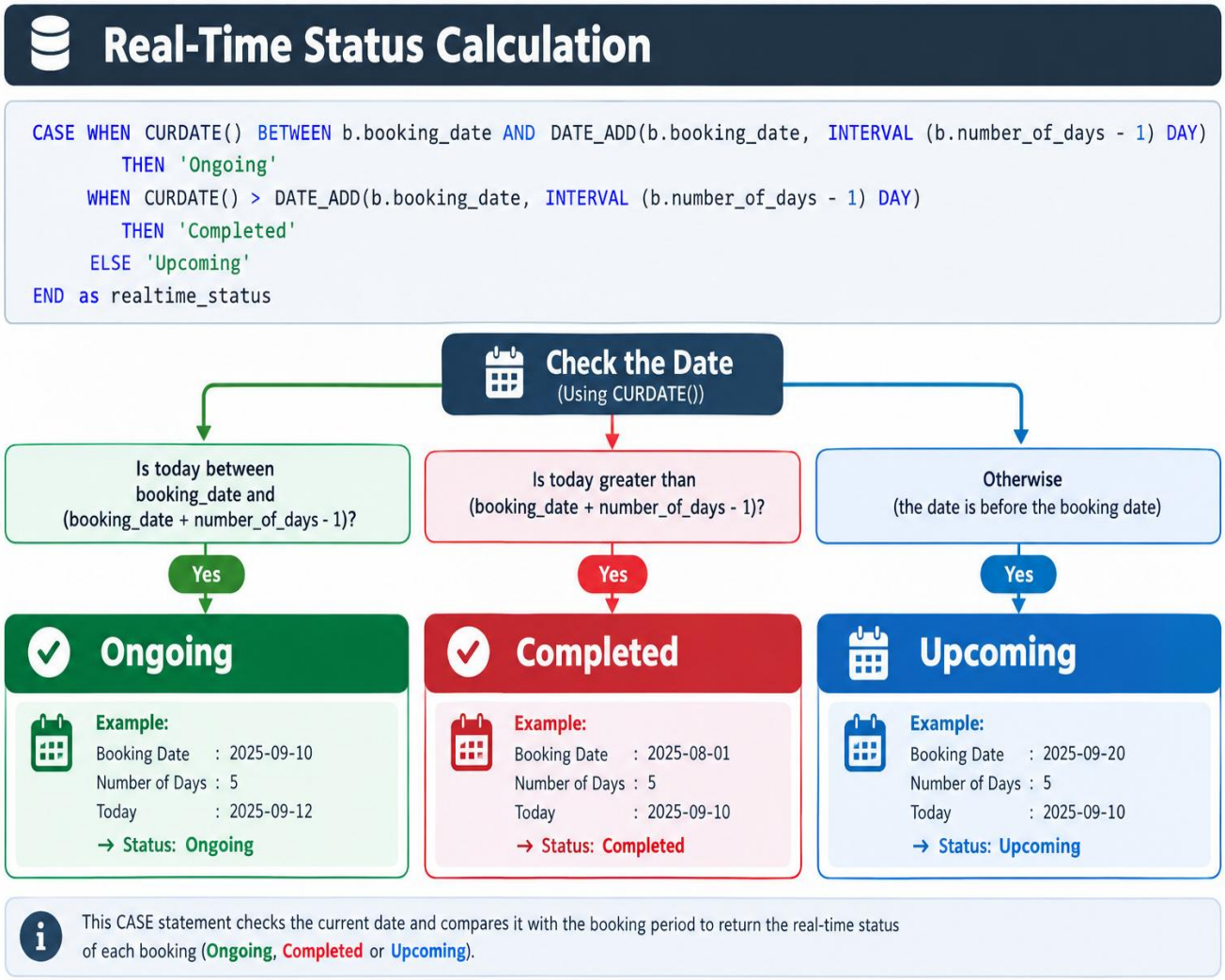
File 2: admin/advanced_reports.php

Purpose: 9-tab business intelligence report.

Tabs Explained:

- **Tab 1 — Profit:** Today/Month/Year revenue, AOV, RPU
- **Tab 2 — Completed Trips:** All completed trips with days ago
- **Tab 3 — Ongoing Trips:** Active trips with progress bars
- **Tab 4 — All Bookings:** Complete list of bookings
- **Tab 5 — Travellers:** All customers with stats
- **Tab 6 — Vehicles:** Vehicle + Driver + Operator performance
- **Tab 7 — Unused:** Packages with no bookings
- **Tab 8 — Deleted:** Soft-deleted records
- **Tab 9 — Top Lists:** Top travellers, top packages, performance scores

Real-Time Status Calculation:



File 3: admin/booking.php

Purpose: Manage all customer bookings.

Key Features:

- Real-time status for each booking
- Progress bars for ongoing trips
- Email confirmation system (PHPMailer)
- DataTables with search/sort/export
- Modal for detailed view

Auto-Confirm Logic: When admin opens this page, the system:

1. Finds all bookings with status = 'pending'
2. Changes them to 'Approved'
3. Sends confirmation email to each customer
4. Shows success message once per session

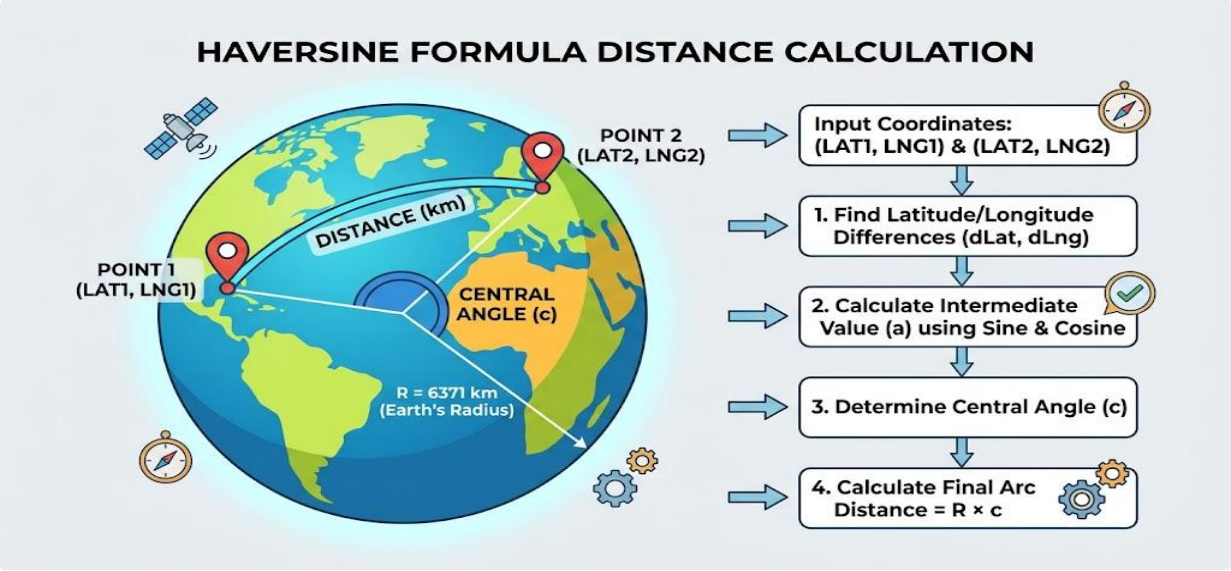
File 4: admin/packages.php

Purpose: Manage tour packages (Create, Read, Update, Delete).

Key Features:

- Card-based grid display
- Image upload (file or URL)
- Google Maps route visualization
- Automatic price calculation using Haversine formula
- Search functionality
- Modal-based CRUD

Haversine Formula for Distance:



File 5: admin/vehicles.php

Purpose: Manage the complete vehicle fleet.

Key Features: Vehicle CRUD, status change dropdown, GPS display, driver assignment, fleet charts

File 6: admin/users.php

Purpose: Manage customer accounts.

Key Features: User list, add user, delete user, view bookings, clear all

File 7: admin/view_feedbacks.php

Purpose: View and reply to customer feedback.

Key Features: List feedback, inline reply form, email reply via PHPMailer, delete

File 8: admin/create_admin.php

Purpose: Create new admin accounts.

File 9: admin/create_vehicle_manager.php

Purpose: Create new vehicle manager accounts (SHA1 password).

File 10: admin/edit_booking.php

Purpose: Edit an existing booking with Google Maps editor.

File 11: admin/delete_booking.php

Purpose: Soft-delete a booking (change status to "Deleted").

File 12: admin/assign_vehicle.php

Purpose: Assign a vehicle to a booking with availability check.

File 13: admin/get_vehicle_locations.php

Purpose: API endpoint returning vehicle GPS data as JSON.

File 14: admin/simulate_gps.php

Purpose: Simulate GPS movements for testing.

File 15: admin/update_location.php

Purpose: API endpoint to update vehicle location (with API key).

File 16: admin/vehicle_tracking.php

Purpose: Show all vehicles on a live Leaflet.js map (auto-refresh 2s).

3.6.2 USER MODULE — 9 Files

File 1: user/dashboard.php

Purpose: Customer's personal dashboard with 4 KPI cards, recent activity, 2 charts.

File 2: user/packages.php

Purpose: Browse packages with card grid, search, sort, and "Book Now" button (with email).

File 3: user/my_bookings.php

Purpose: View/manage bookings with real-time status, filters, confetti animation.

File 4: user/payments.php

Purpose: Make payments with card formatting, camera option, history table.

File 5: user/plan_a_trip.php

Purpose: Custom trip planner with Google Maps, distance & cost calculation, group discounts.

File 6: user/feedback_status.php

Purpose: View feedback status and reply to admin.

File 7: user/view_package.php

Purpose: Show full package details with route map and social share.

File 8: user/edit_booking.php

Purpose: Edit booking with Leaflet map and reverse geocoding.

File 9: user/delete_booking.php

Purpose: Delete a booking with animated progress display.

3.6.3 VEHICLE MANAGER MODULE — 12 Files

File 1: vehicle_management/vehicle_manager_login.php

Purpose: Login for vehicle managers (SHA1 password verification).

File 2: vehicle_management/dashboard.php

Purpose: Manager's main dashboard with 6 KPI cards, revenue chart, activity feed, and real-time vehicle release.

File 3: vehicle_management/vehicles.php

Purpose: Full vehicle management with inline edit, bulk update, driver assignment, charts.

File 4: vehicle_management/add_vehicle.php

Purpose: Add new vehicle with live preview, draft saving, clone feature, duplicate check.

File 5: vehicle_management/update.php

Purpose: Update vehicles (single, bulk, inline) with audit logging.

File 6: vehicle_management/filter.php

Purpose: Advanced vehicle filter with comparison, operator tabs, date range.

File 7: vehicle_management/drivers.php

Purpose: Complete driver management with assign/unassign.

File 8: vehicle_management/reports.php

Purpose: Advanced reporting with Kanban drag-drop board and table view.

File 9: vehicle_management/delete_vehicle.php

Purpose: Permanently delete a vehicle.

File 10: vehicle_management/database.php

Purpose: Database connection helper.

File 11: vehicle_management/get_vehicle_calendar.php

Purpose: Get vehicle booking calendar data as HTML table.

File 12: vehicle_management/get_vehicle_history.php

Purpose: Get vehicle audit history.

3.7 Function Library (Every Function Explained)

Function	Location	Purpose
getValue()	admin/dashboard.php	Get a single value from a query
getRow()	admin/dashboard.php	Get one complete row
getRows()	admin/dashboard.php	Get multiple rows

autoConfirmExistingBookings()	admin/booking.php	Auto-approve pending bookings
sendConfirmationEmail()	admin/booking.php	Send booking email
sendManualEmail()	admin/booking.php	Manually send email
readAll()	admin/booking.php	Fetch all bookings with status
calculateAutoPrice()	admin/packages.php	Calculate price from distance
togglePasswordVisibility()	Multiple	Show/hide password
filterPackages()	admin/packages.php	Filter packages by search
updateBookings()	admin/booking.php	Live update from AJAX
animateNumbers()	Multiple	Animate KPI counters
initCharts()	Multiple	Initialize Chart.js
initClock()	Multiple	Live time display
initTheme()	Multiple	Dark/light toggle
createParticles()	Multiple	Floating particles
openTab()	advanced_reports.php	Switch report tabs

exportPDF()	advanced_reports.php	Export PDF report
filterTable()	user/my_bookings.php	Filter bookings
sortTable()	user/my_bookings.php	Sort bookings
exportCSV()	Multiple	Export to CSV
copyTxn()	user/payments.php	Copy transaction ID
launchConfetti()	Multiple	Confetti animation
calculateDistance()	plan_a_trip.php	Distance calculation
updateAllCalculations()	plan_a_trip.php	Recalculate costs
openMapModal()	Multiple	Open route map
showToast()	Multiple	Show notification
togglePassword()	Multiple	Password show/hide

3.8 JavaScript Functions (Frontend Behavior)

UI Functions:

- **initTheme()** — Dark/light theme toggle with localStorage
- **initClock()** — Live clock updating every 1 second
- **createParticles()** — Creates 30-40 floating dots
- **animateNumbers()** — Animates KPI cards from 0 to target
- **typeWriter()** — Types heading text character by character
- **ripple Effect** — Expands circle from click point

Chart Functions:

- **initCharts()** — Creates Bar, Line, Doughnut, Pie charts

Data Functions:

- **filterTable()** — Filters rows by search input
- **sortTable()** — Sorts visible rows
- **exportCSV()** — Builds CSV from visible rows

Modal Functions:

- **openModal(data)** — Shows detail modal
- **closeModal()** — Hides modal
- **openMapModal()** — Initializes Google Map

Interactive Functions:

- **onDragStart/onDrop/onDragOver/onDragEnd** — Kanban drag-drop
- **togglePasswordVisibility()** — Show/hide password
- **showToast()** — Show notification

3.9 User Workflows (Step by Step)

Workflow 1: Customer Books a Tour Package

1. Customer logs in at index.php → redirected to user/dashboard.php
2. Customer browses packages in user/packages.php
3. Customer clicks "Book Now" → System sends GET request
4. PHP fetches package, inserts booking with status "Approved"
5. PHPMailer sends confirmation email automatically
6. Customer sees confetti animation and success message
7. Booking shows in my_bookings.php with "Upcoming" status
8. On trip day, status becomes "Ongoing" with progress bar
9. Trip completes → status becomes "Completed"

Workflow 2: Customer Plans Custom Trip

1. Customer navigates to plan_a_trip.php
2. Clicks on map to place pickup marker (P)
3. Clicks again to place destination marker (D)
4. System calculates distance (Google Directions API)
5. Calculates empty leg + return leg distance
6. Customer selects vehicle from dropdown
7. Sets days (1-30) and members (1-vehicle capacity)
8. System calculates: package cost + vehicle cost + petrol cost
9. Group discount applied (10%-20% based on members)
10. Customer clicks "Finalize Order" → System checks availability
11. If available, inserts booking + sends email

Workflow 3: Admin Manages Bookings

1. Admin logs in → navigates to booking.php
2. System auto-confirms pending bookings + sends emails
3. Admin views booking table with real-time status
4. Clicks booking row → Modal opens with full details
5. Admin can: send email, edit, delete, or view route map

Workflow 4: Admin Generates Reports

1. Admin navigates to advanced_reports.php
2. Selects date range (From / To)
3. Clicks "Apply" → System filters data

4. Admin views 9 tabs of reports
5. Clicks "Print" or "PDF" for any tab

Workflow 5: Vehicle Manager Updates Vehicle

1. Manager logs in → redirected to dashboard.php
2. System releases finished vehicles automatically
3. Manager navigates to vehicles.php
4. Manager clicks editable field → Prompt appears
5. Manager enters new value → AJAX saves to database
6. Success toast appears

Workflow 6: Manager Uses Kanban Board

1. Manager navigates to reports.php
2. Sees 3 columns: Upcoming, Ongoing, Completed
3. Drags a card from one column to another
4. System updates database via AJAX
5. Shows toast notification
6. Column counts update automatically

Workflow 7: GPS Simulation and Tracking

1. Admin navigates to simulate_gps.php
2. Clicks "Update" for one vehicle (or "Update All")
3. Random GPS coordinates generated and saved
4. Admin navigates to vehicle_tracking.php
5. Sees Leaflet map with color-coded markers
6. Map auto-refreshes every 2 seconds

Workflow 8: Password Reset

1. User clicks "Forgot Password" → forgot_password.php
2. Enters email → System checks all 3 tables
3. If found, generates token + sends email
4. User clicks link → reset_password.php?token=XYZ
5. System validates token + expiry
6. User enters new password → Updates all 3 tables
7. Token deleted → User can login with new password

3.10 Security Features

1. **Authentication:** Session-based login with 3 different tables for 3 roles
2. **Password Hashing:** Users (bcrypt), Admin (plain text), Vehicle Manager (SHA1)
3. **SQL Injection Prevention:** Uses mysqli_real_escape_string() and prepare()
4. **XSS Prevention:** Uses htmlspecialchars() for output
5. **CSRF Protection:** Token generated and verified on form submit
6. **Session Security:** Cookie deletion, cache-control headers
7. **API Key Check:** update_location.php requires API key
8. **Input Validation:** Email, phone, price, member count
9. **Audit Logging:** Every important action logged
10. **Soft Delete:** Bookings soft-deleted (status = 'Deleted')
11. **File Upload Validation:** Only allows jpg, jpeg, png, gif, webp, bmp
12. **Auto-Complete Protection:** Session-based access control

CHAPTER 04: CONCLUSIONS & FUTURE WORK

4.1 Conclusions

The **Travel & Tour Management System** successfully addresses all the operational problems that small travel agencies face. By replacing manual paper-based processes with a modern web-based system, it improves:

- **Accuracy** — No more double bookings or missed emails
- **Speed** — Bookings confirmed in seconds, not hours
- **Transparency** — Real-time status for all users
- **Profitability** — Detailed analytics to understand the business
- **Customer Satisfaction** — Modern, beautiful interface with instant feedback

The system has **three complete user roles**:

1. **Administrator** — Controls the entire business
2. **Vehicle Manager** — Focuses on fleet operations
3. **Customer** — Books trips and tracks progress

The most innovative feature is the **Real-Time Trip Status Engine** which automatically:

- Calculates if a booking is Upcoming, Ongoing, or Completed
- Uses current date + booking date + number of days
- Updates status without any manual work
- Triggers email confirmations automatically
- Releases vehicles when trips finish

The system uses **free and open-source technologies**, making it affordable for any small travel agency. The glassmorphism UI with animated video background gives it a modern, premium feel.

In summary, the project achieves all its academic objectives and provides a **reliable, scalable, and fully functional** web solution for the modern tourism industry.

4.2 Future Development

Here are improvements we recommend for future versions:

1. **Real Payment Gateway** — Connect to PayHere, Stripe for real card payments
2. **Mobile Application** — Build Android and iOS apps using Flutter or React Native
3. **SMS Notifications** — Send SMS for booking confirmations via Twilio
4. **AI Recommendation Engine** — Suggest packages based on past bookings
5. **Advanced Analytics** — Predictive revenue forecasting, churn prediction
6. **Multi-Language Support** — Sinhala, Tamil, English
7. **Real GPS Device Integration** — Integrate with actual GPS hardware
8. **Live Chat Support** — Chat between customers and admin using Socket.IO
9. **Loyalty Rewards Program** — Points, tiers, referral rewards
10. **Cloud Deployment** — Deploy on AWS or DigitalOcean with auto-scaling
11. **Multi-Tenant Support** — Support multiple travel agencies
12. **Advanced Reporting** — Scheduled email reports, custom builder

CHAPTER 05: REFERENCES

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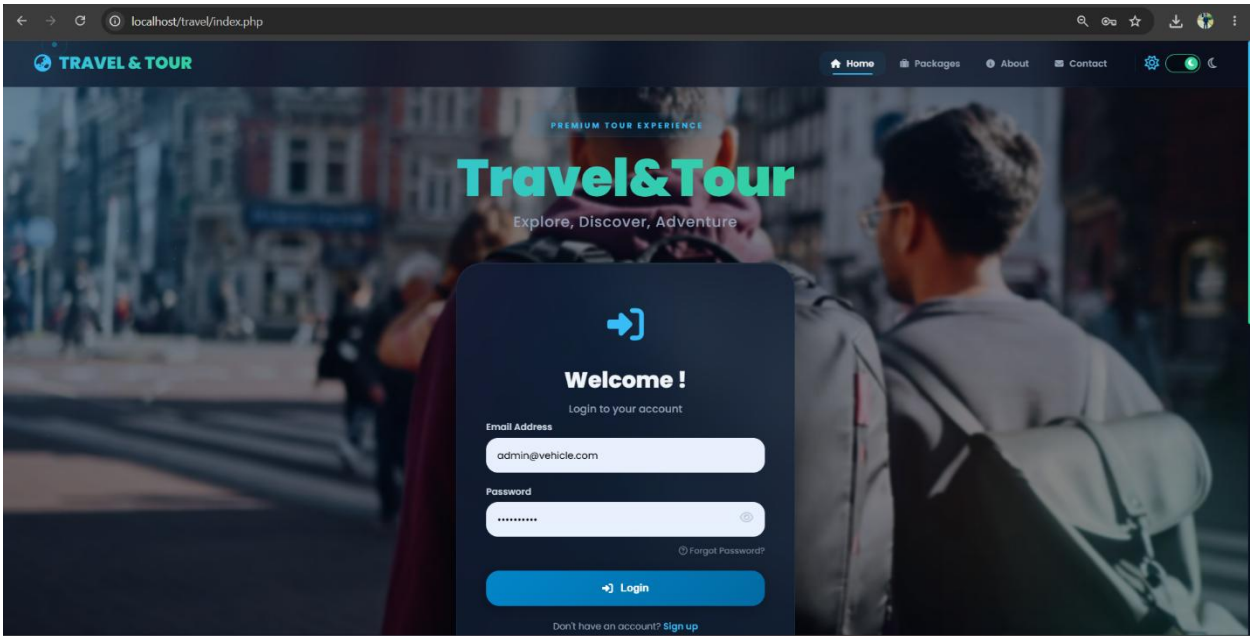
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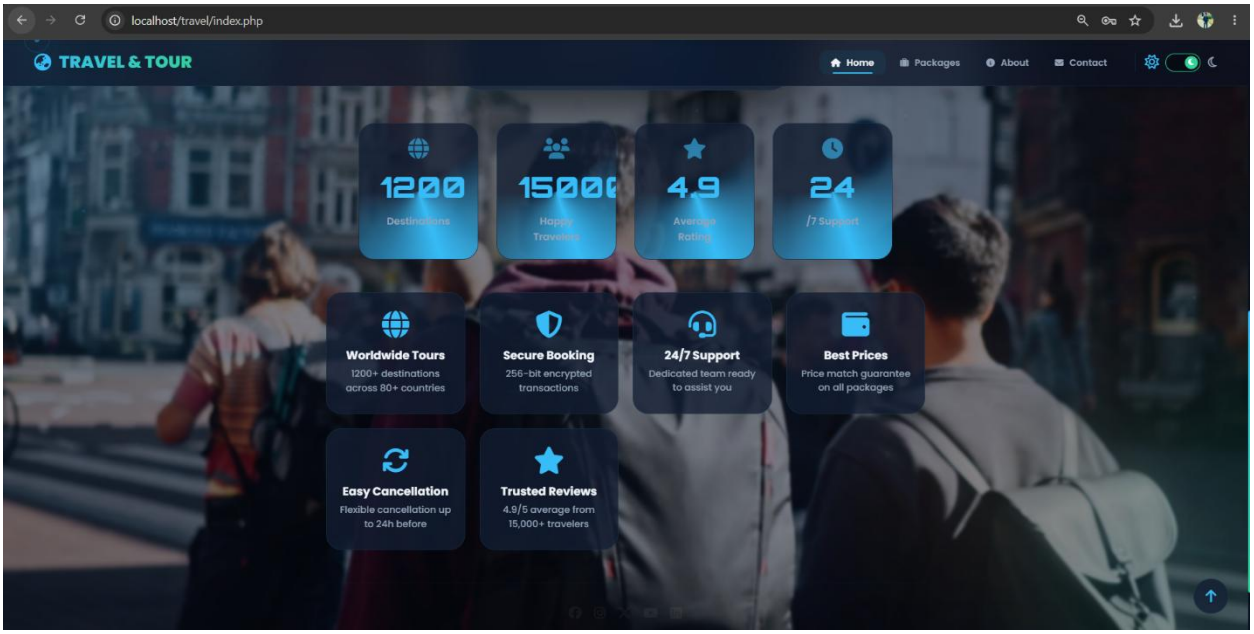
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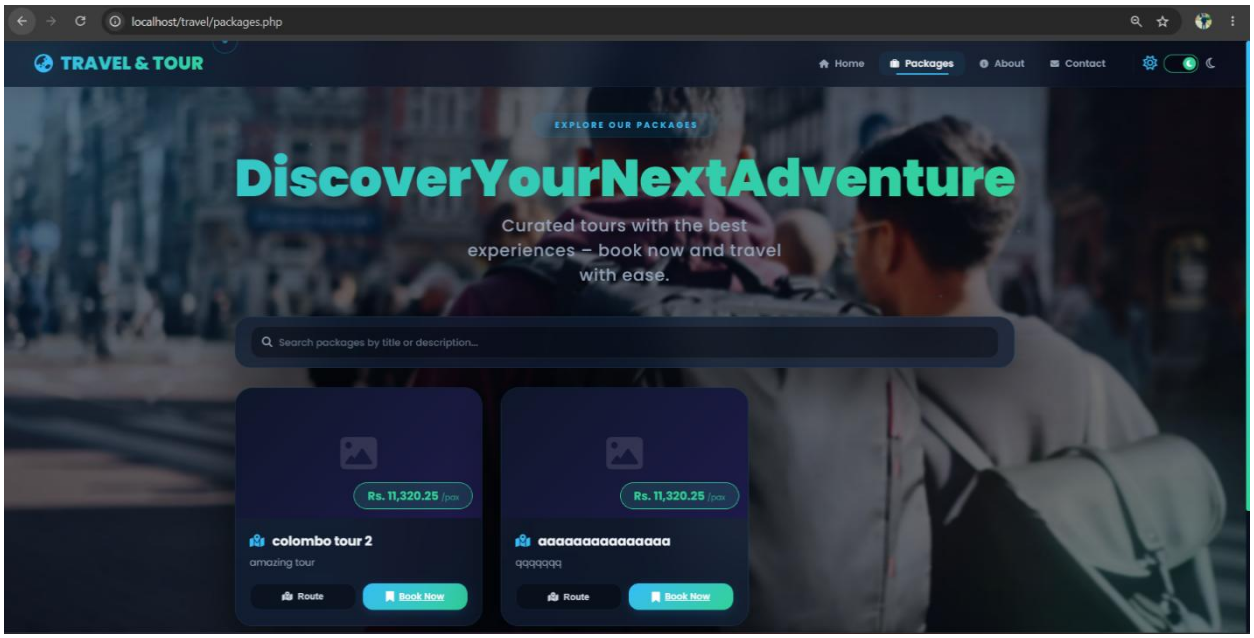
SCREENSHOTS

Figure 1 — Welcome / Login Interface (index.php)

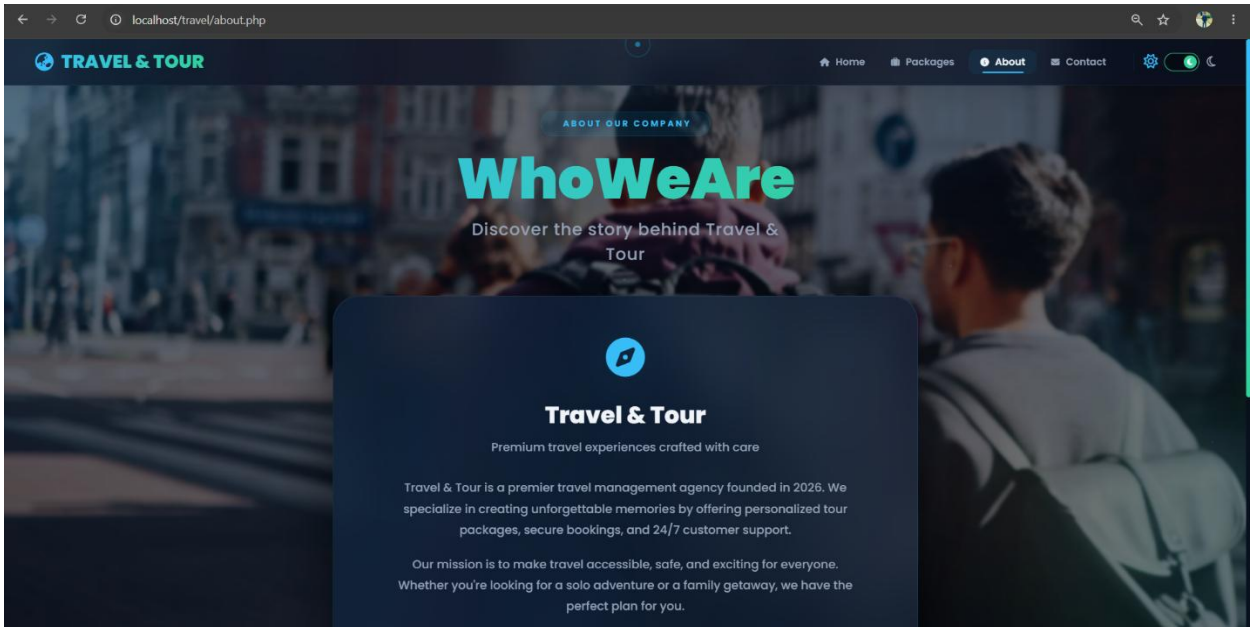




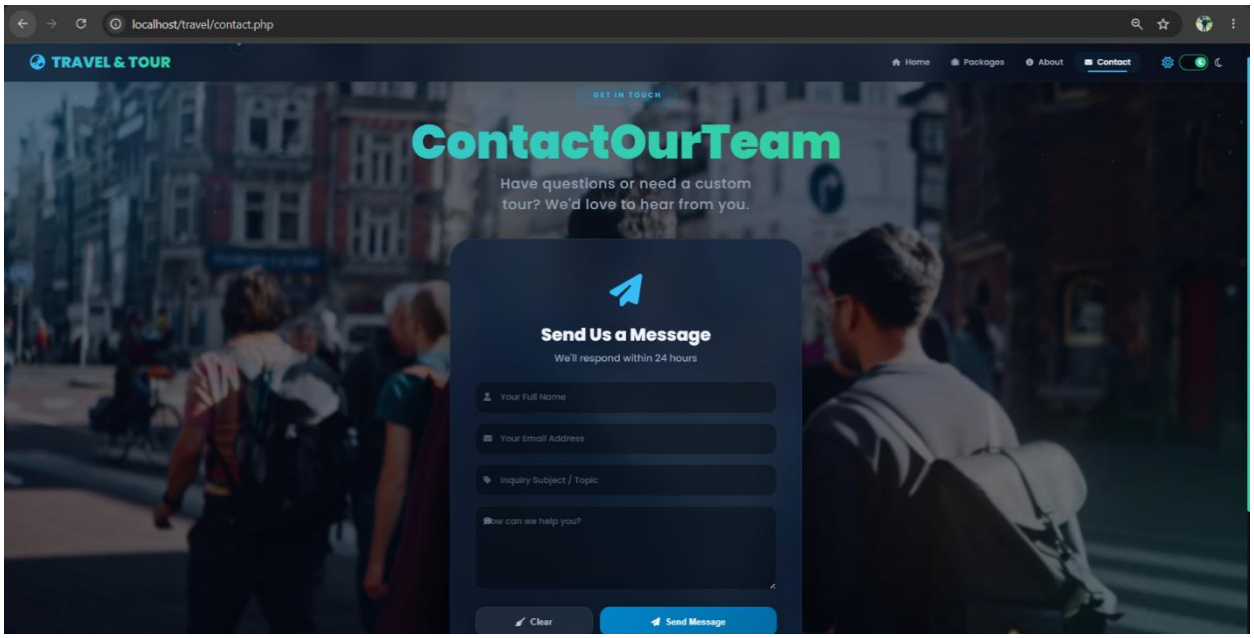
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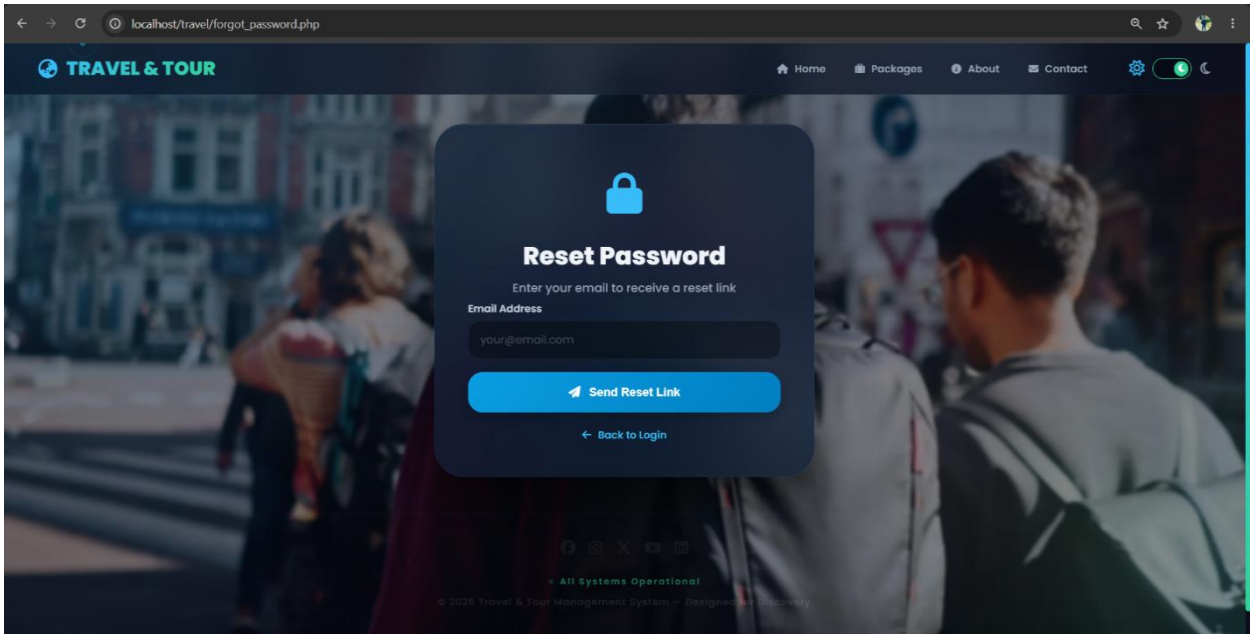
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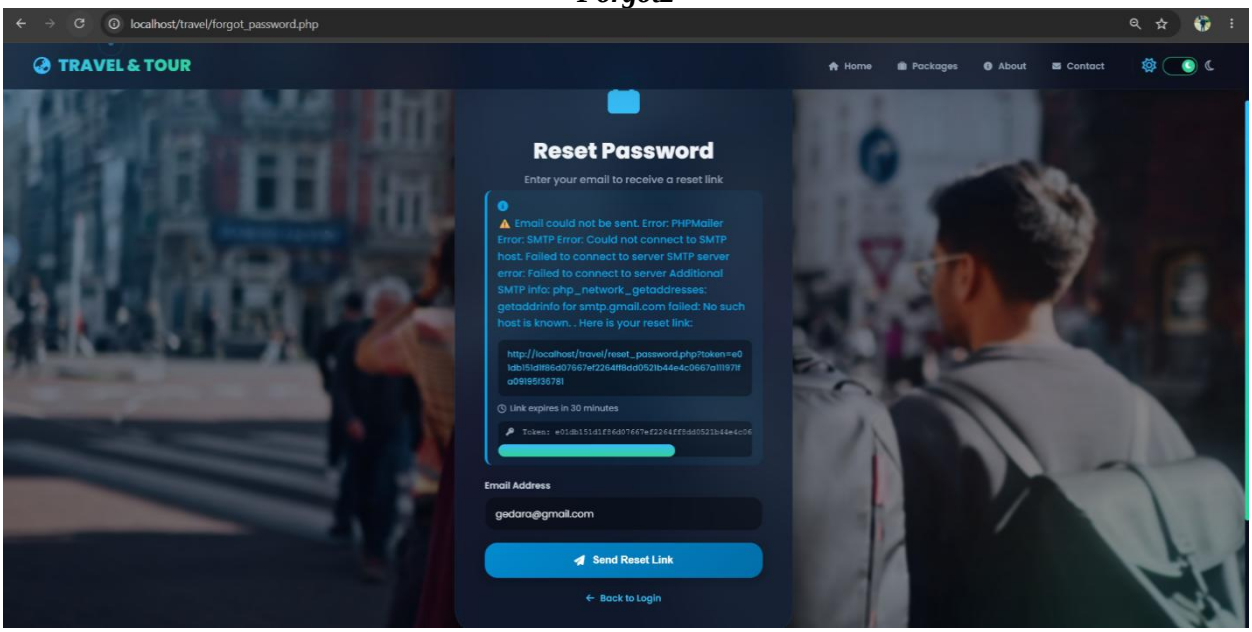
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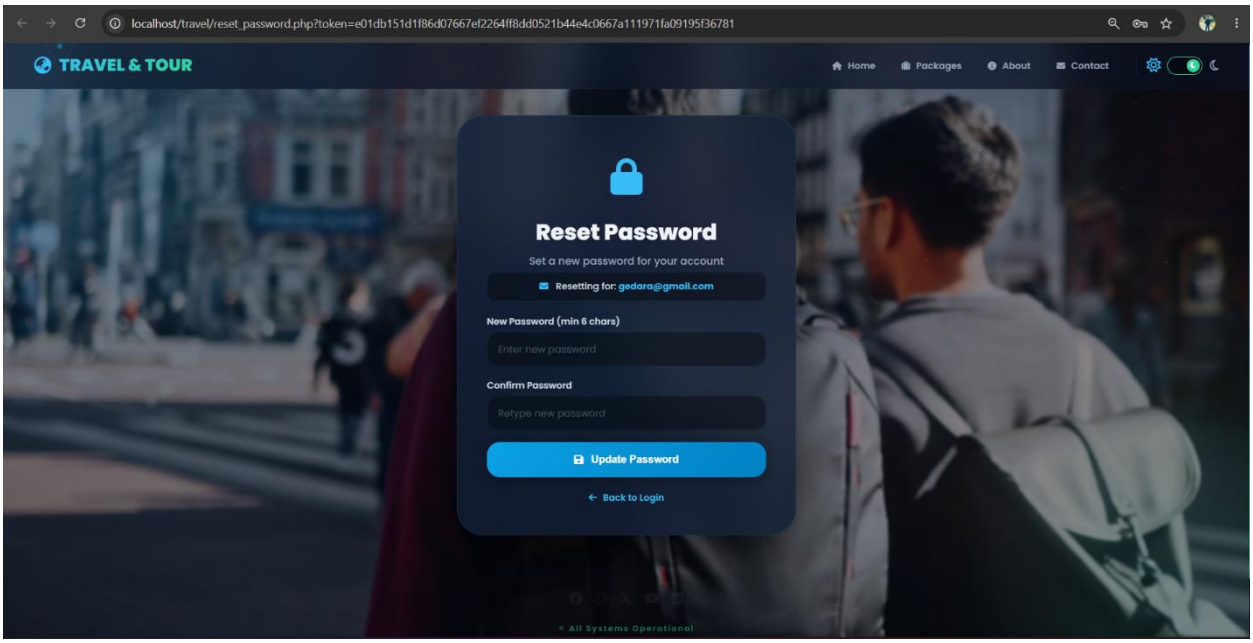
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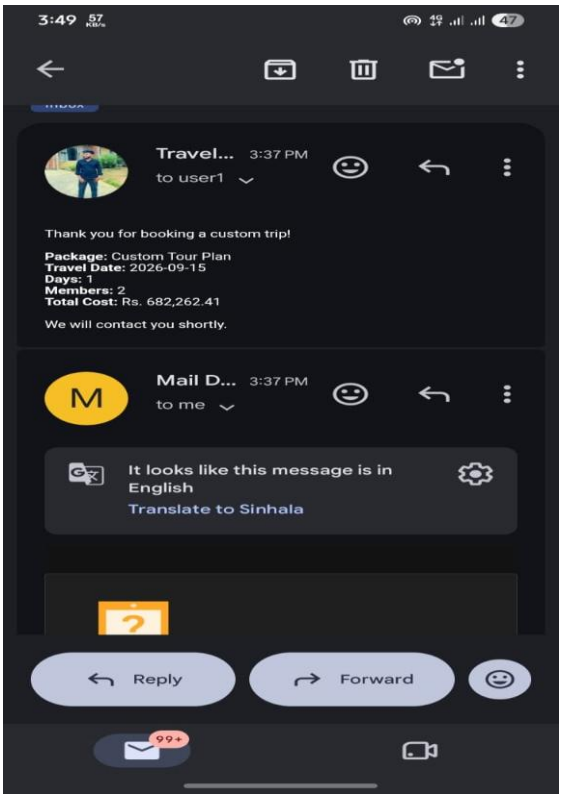
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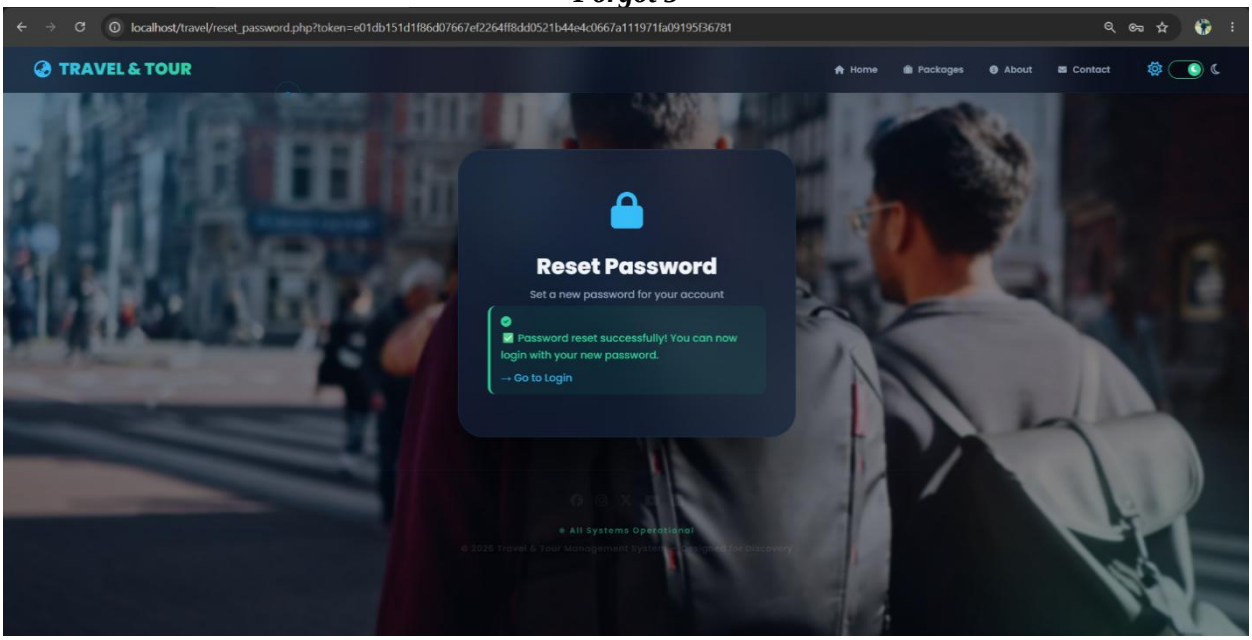
Forgot 3



Forgot4



Forgot 5



Create new user

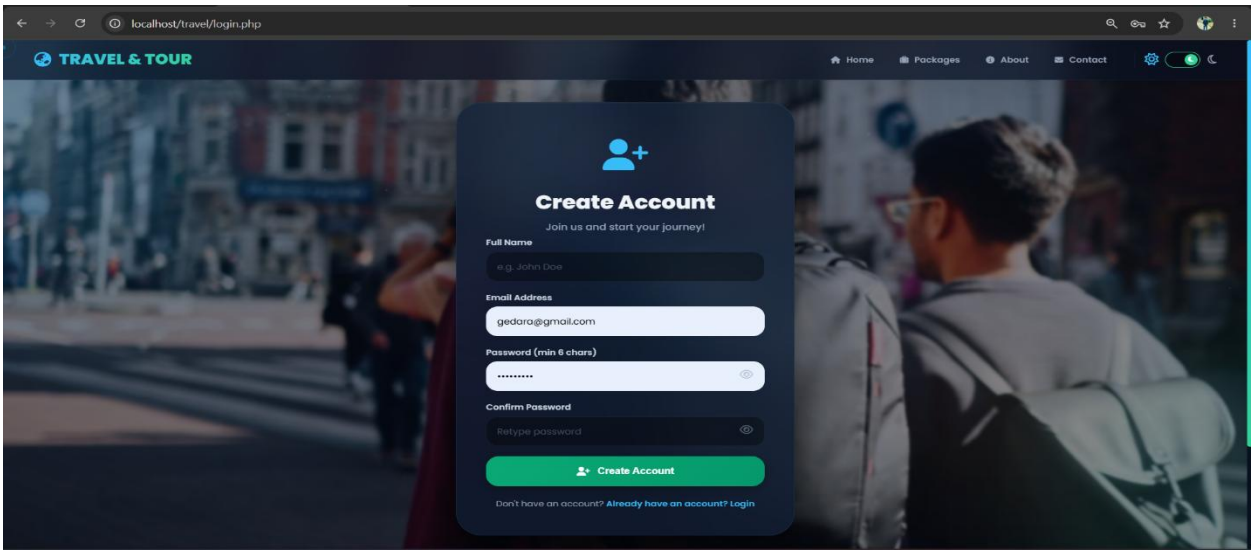
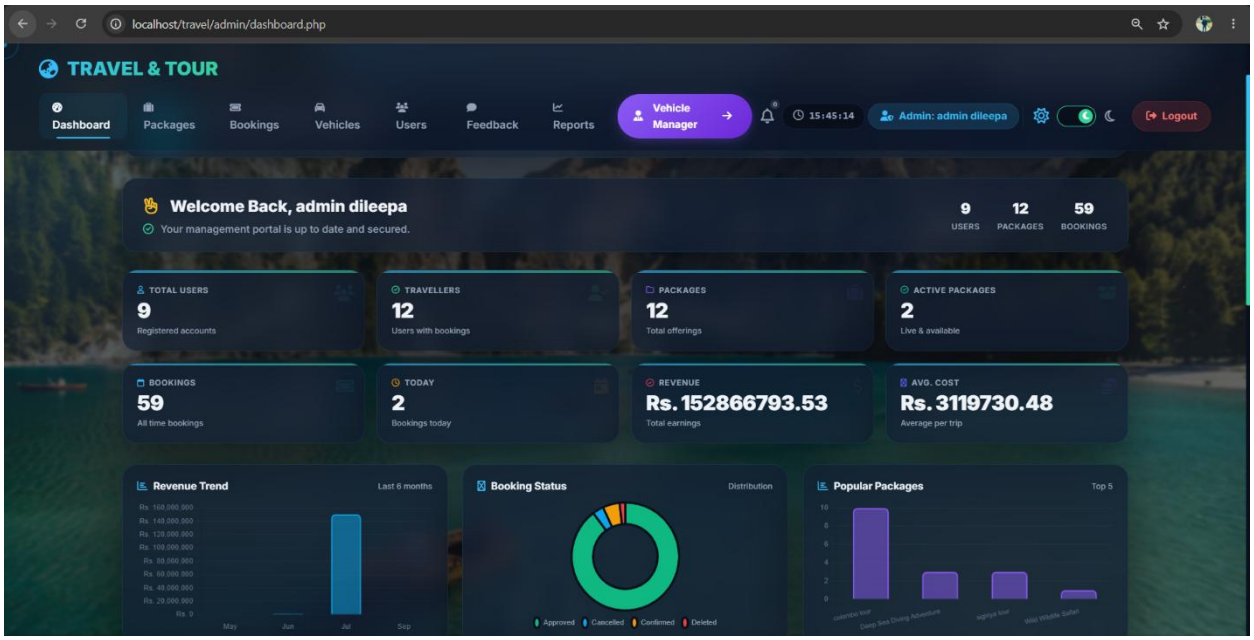


Figure 2 — Admin Dashboard



Dashboard 2

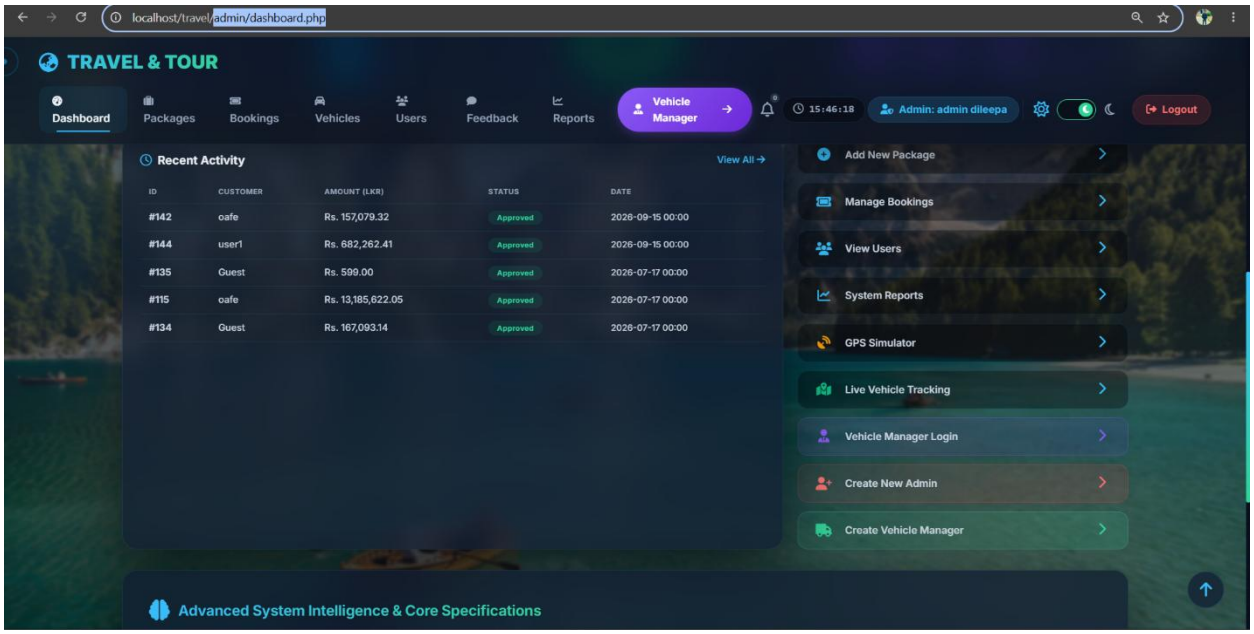
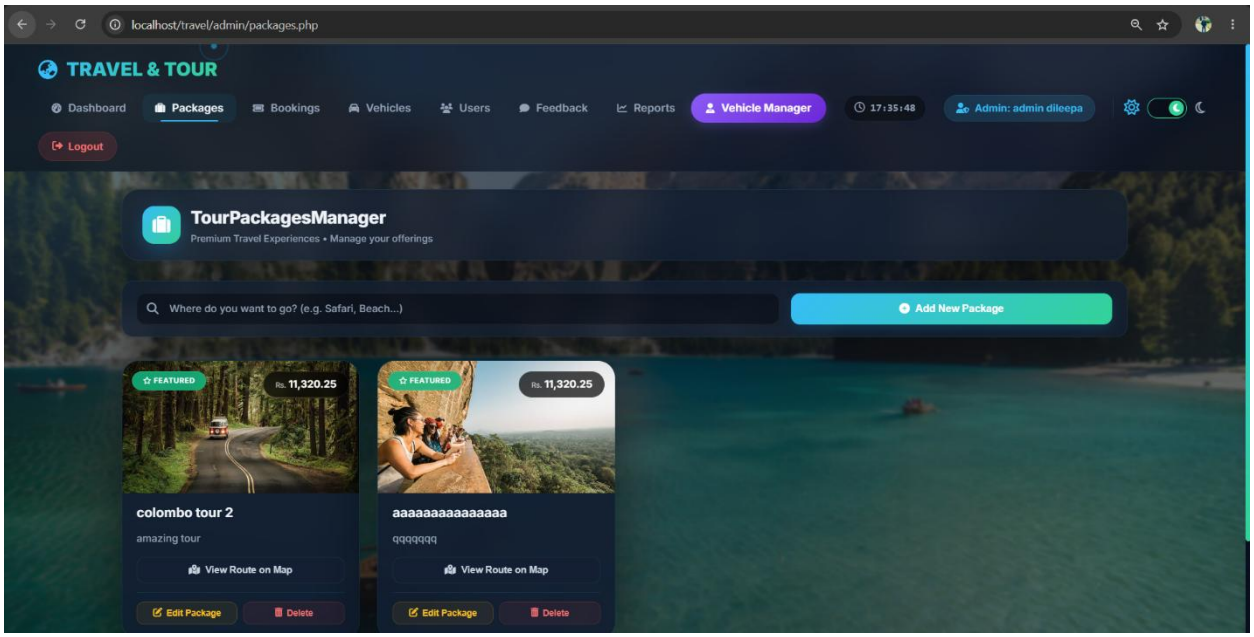


Figure 3 — Admin Package Management



add new Package

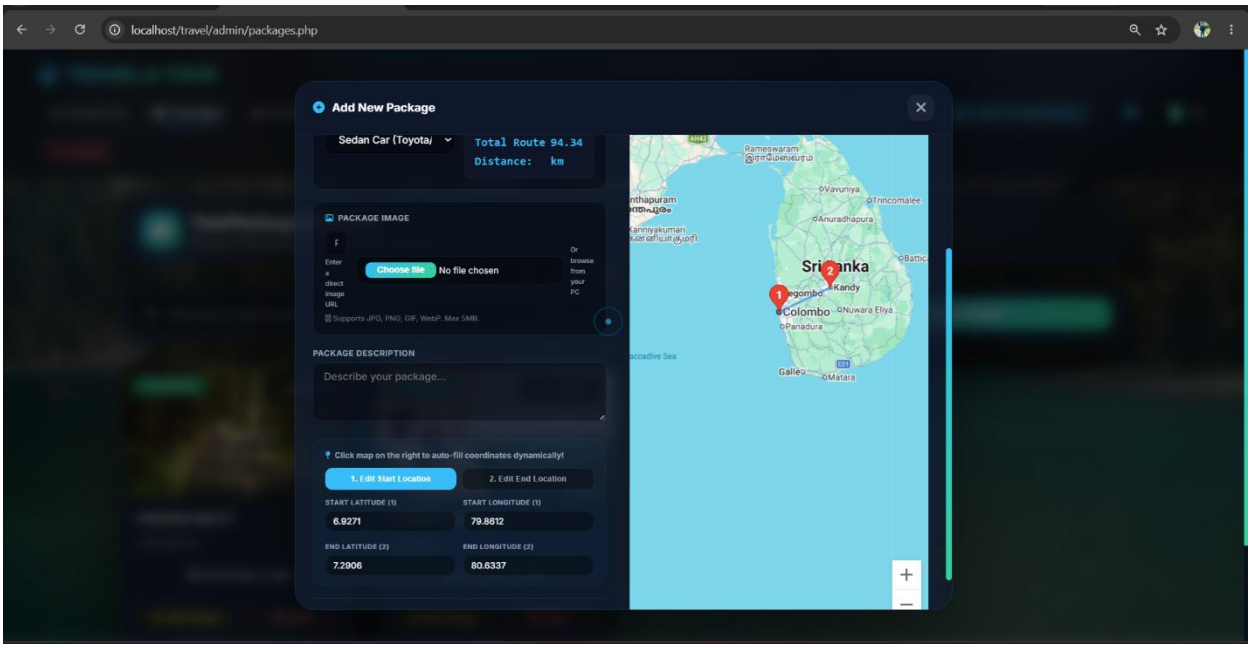
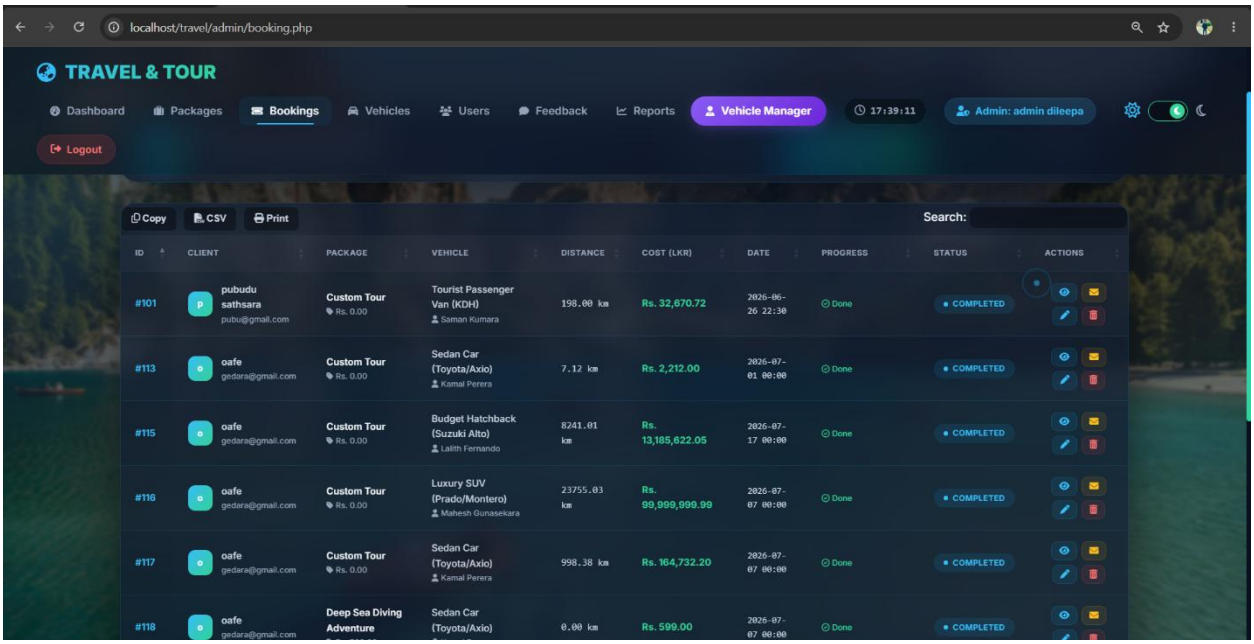
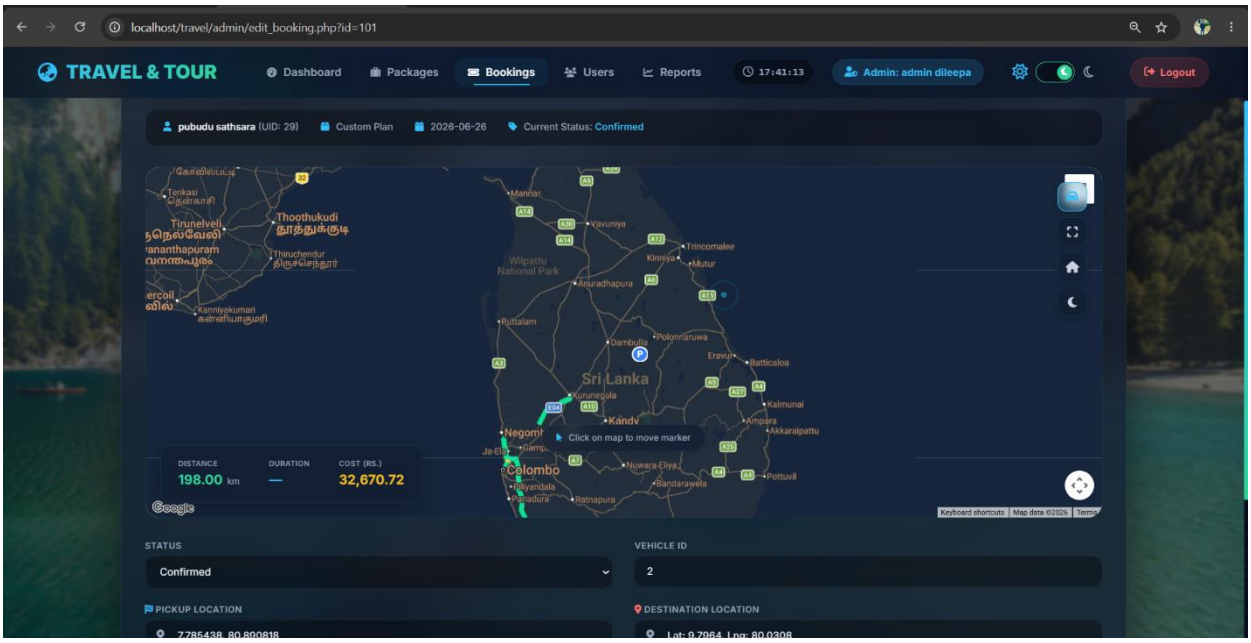


Figure 4 — Admin Booking Management



Admin Edit bookings



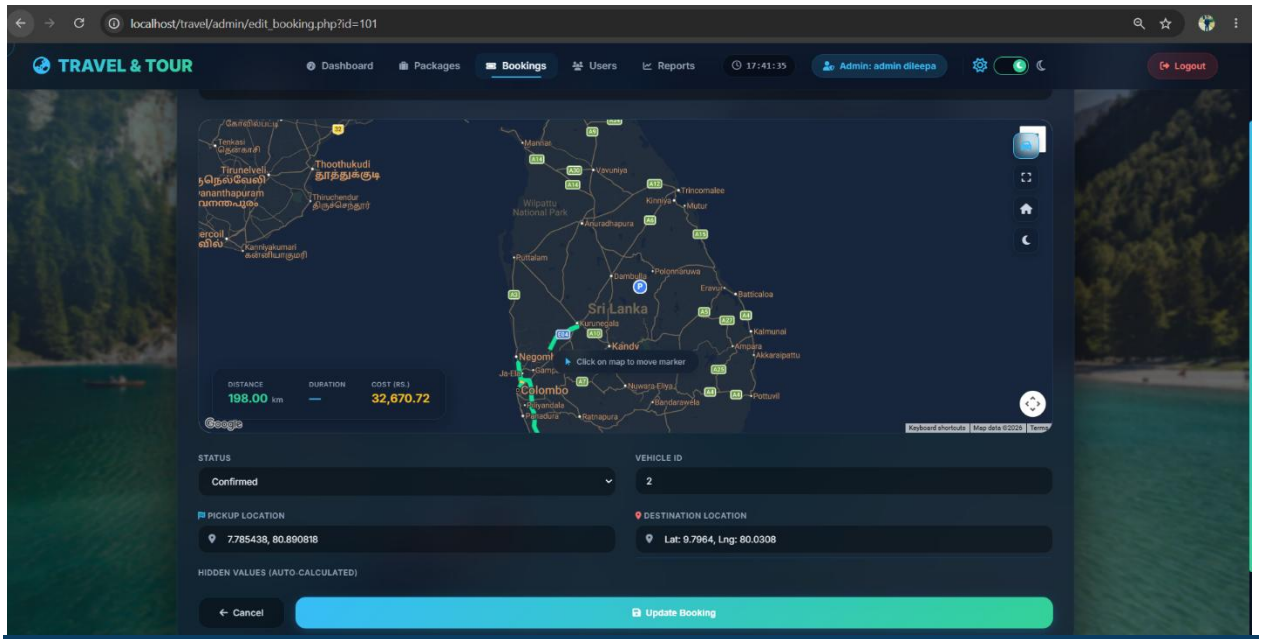


Figure 9 — Admin User Management

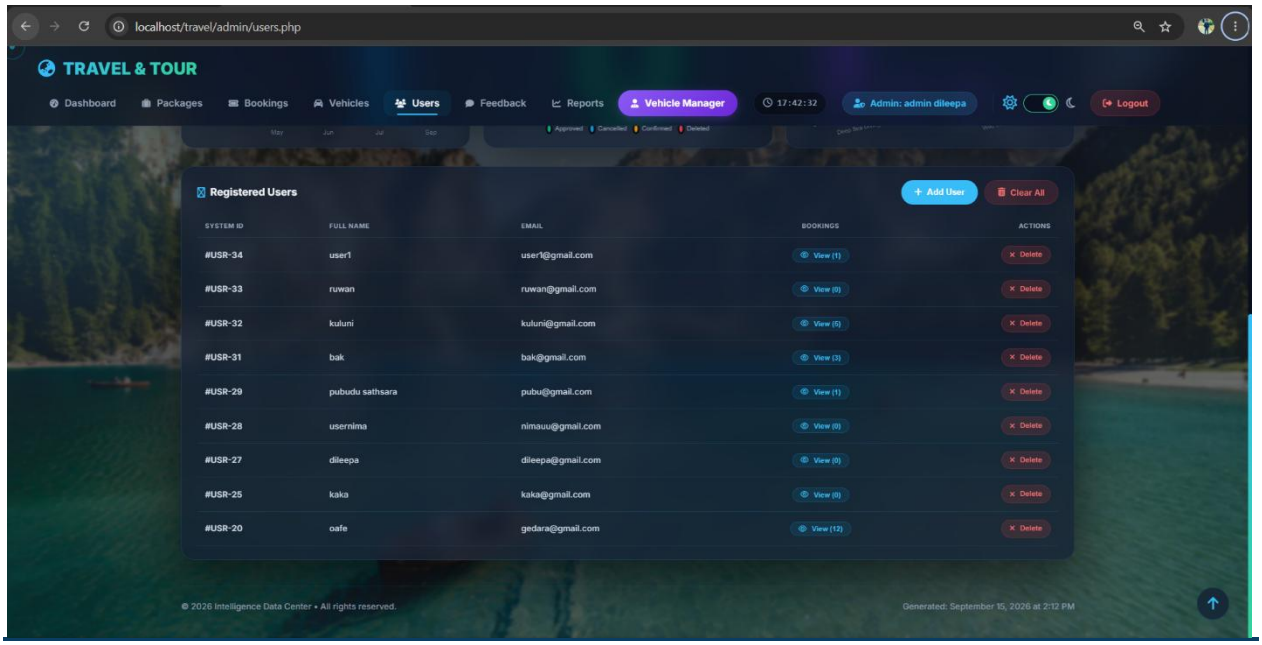


Figure 10 — Admin Feedback Management

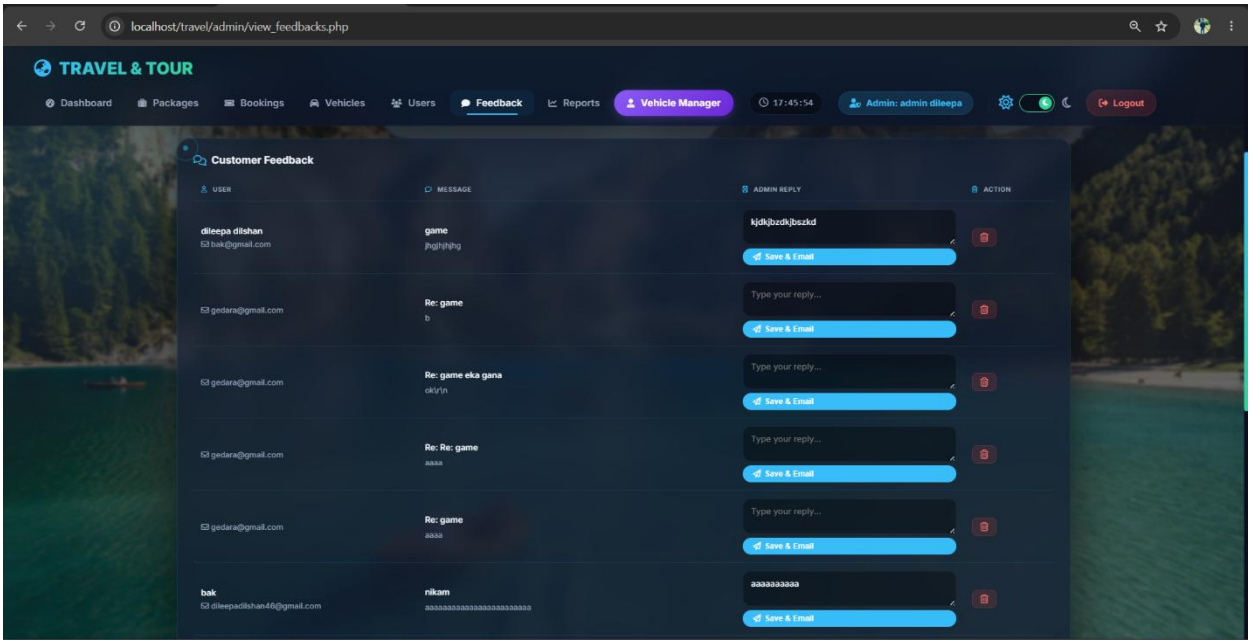
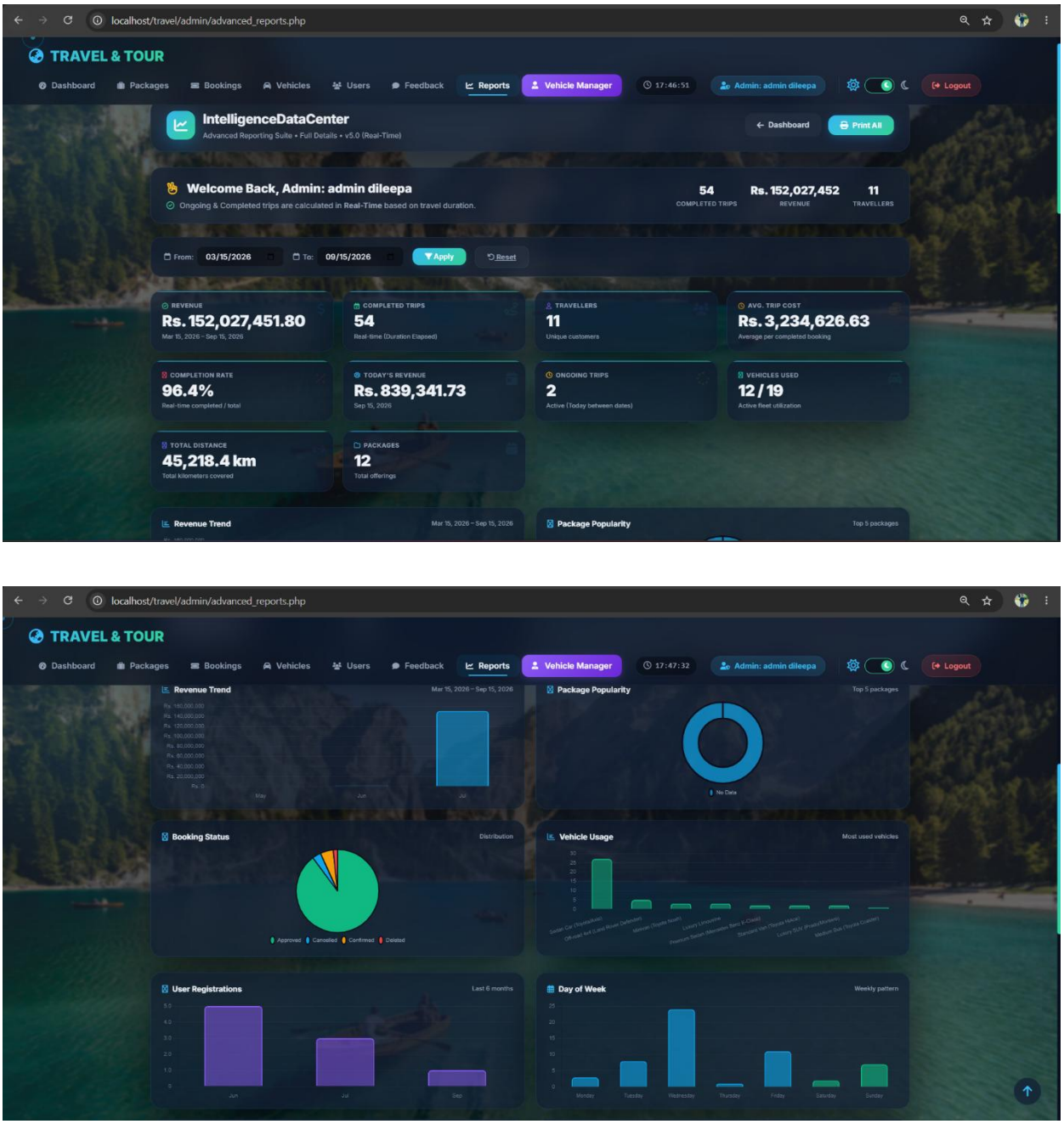


Figure 6 — Admin Advanced Reports Suite



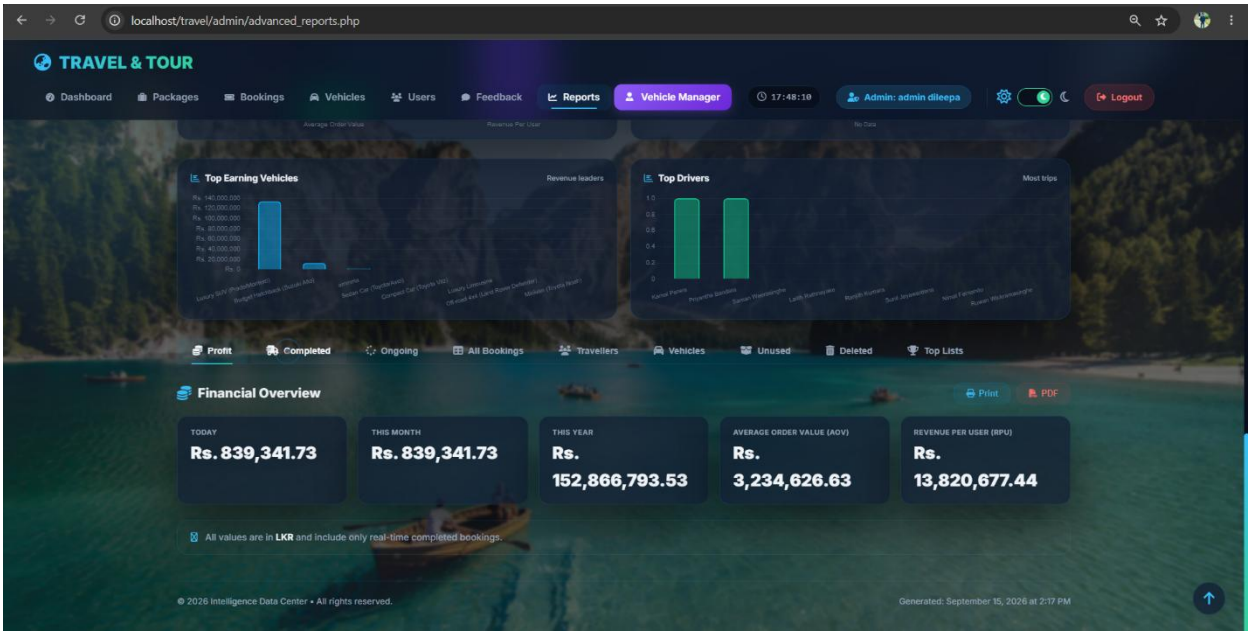


Figure 7 — Admin GPS Simulator

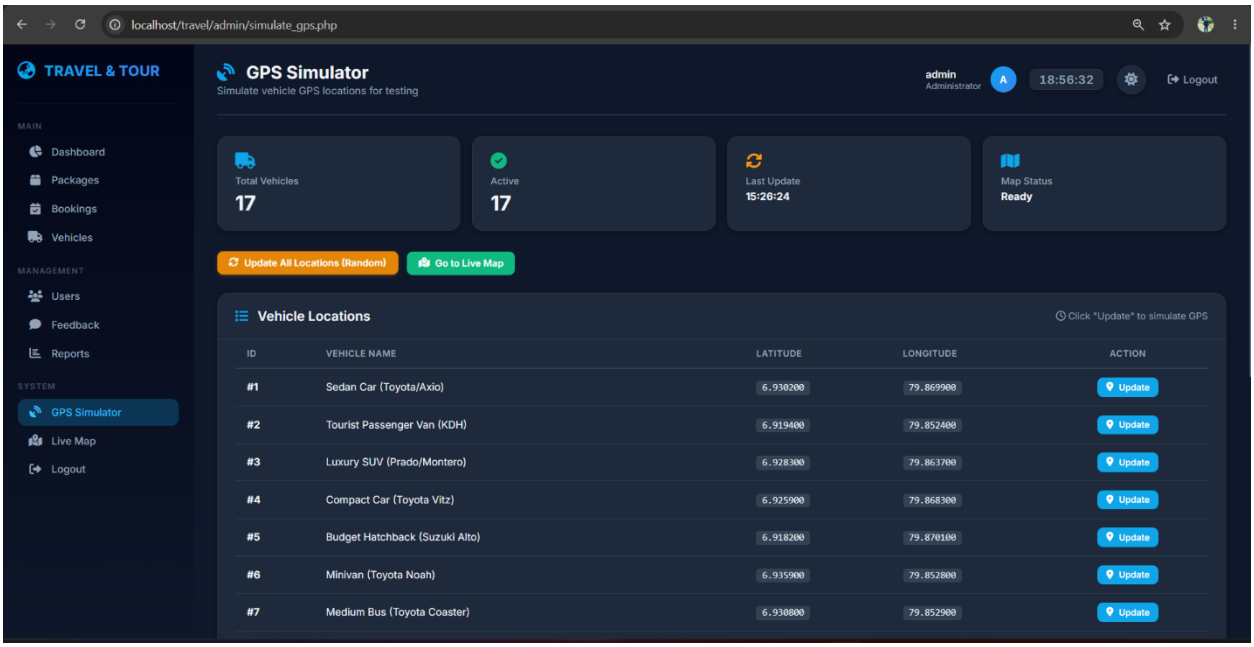


Figure 8 — Admin Live Vehicle Tracking Map

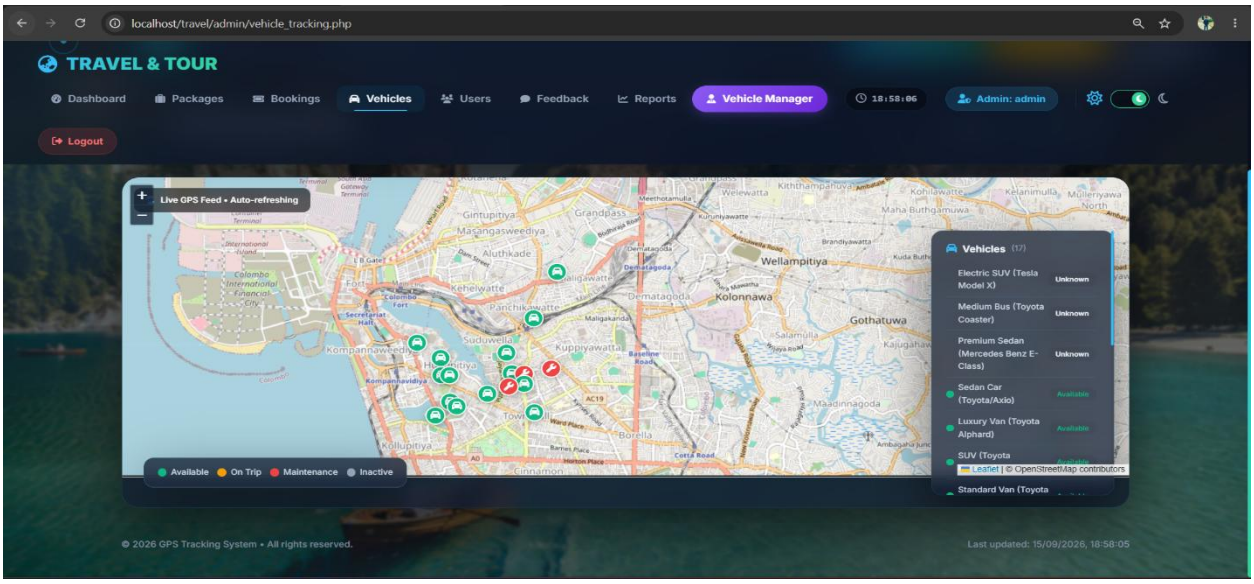


Figure 11 — Vehicle Manager Login

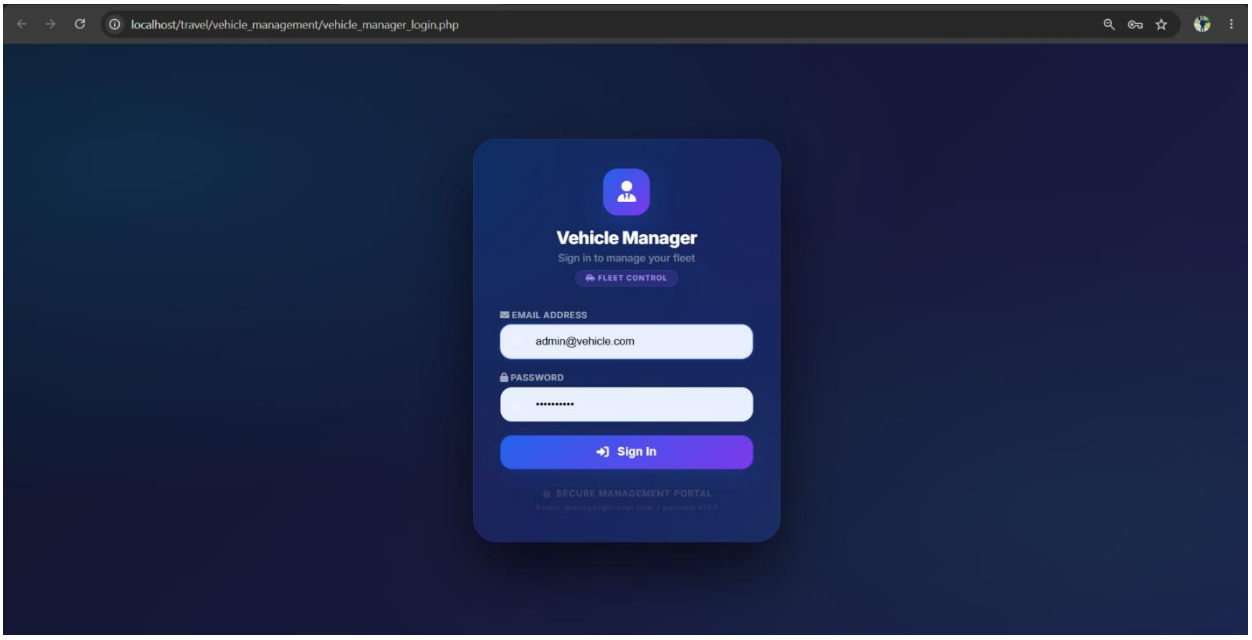
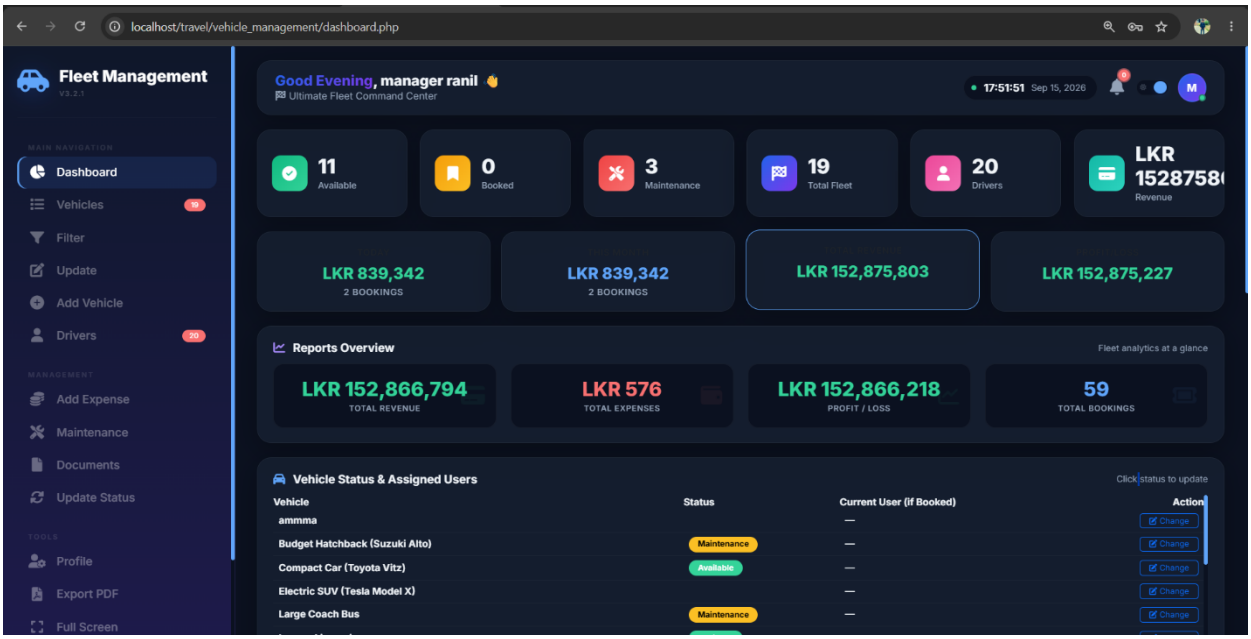
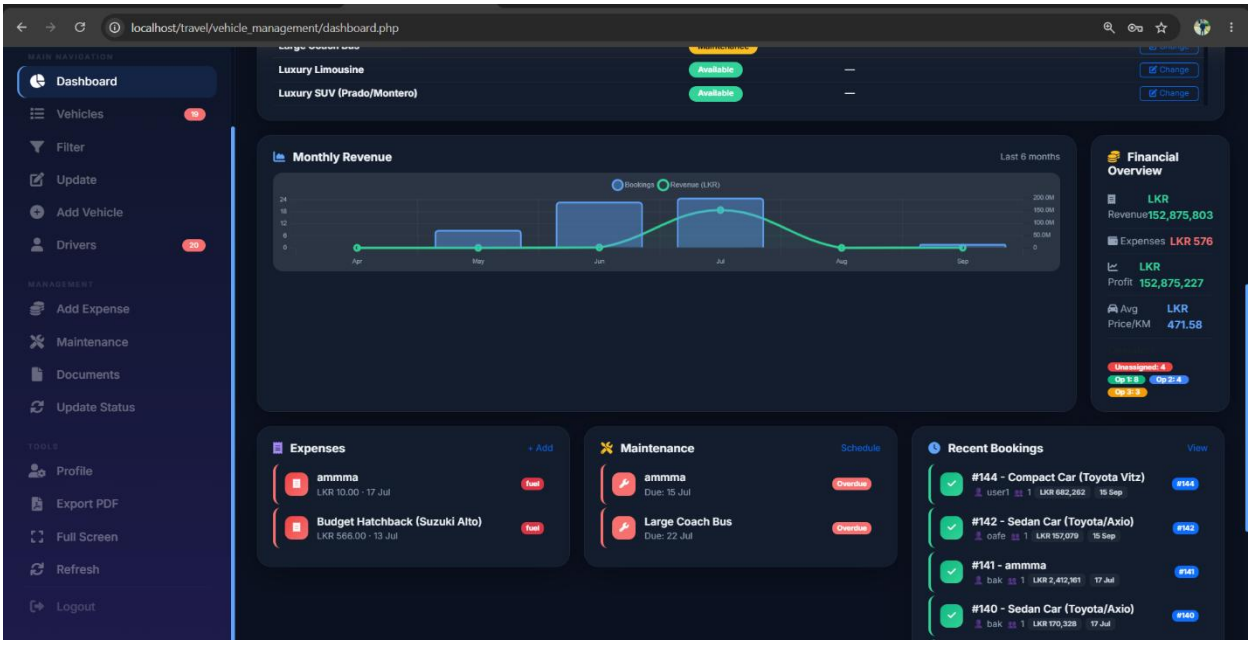


Figure 12 — Vehicle Manager Dashboard





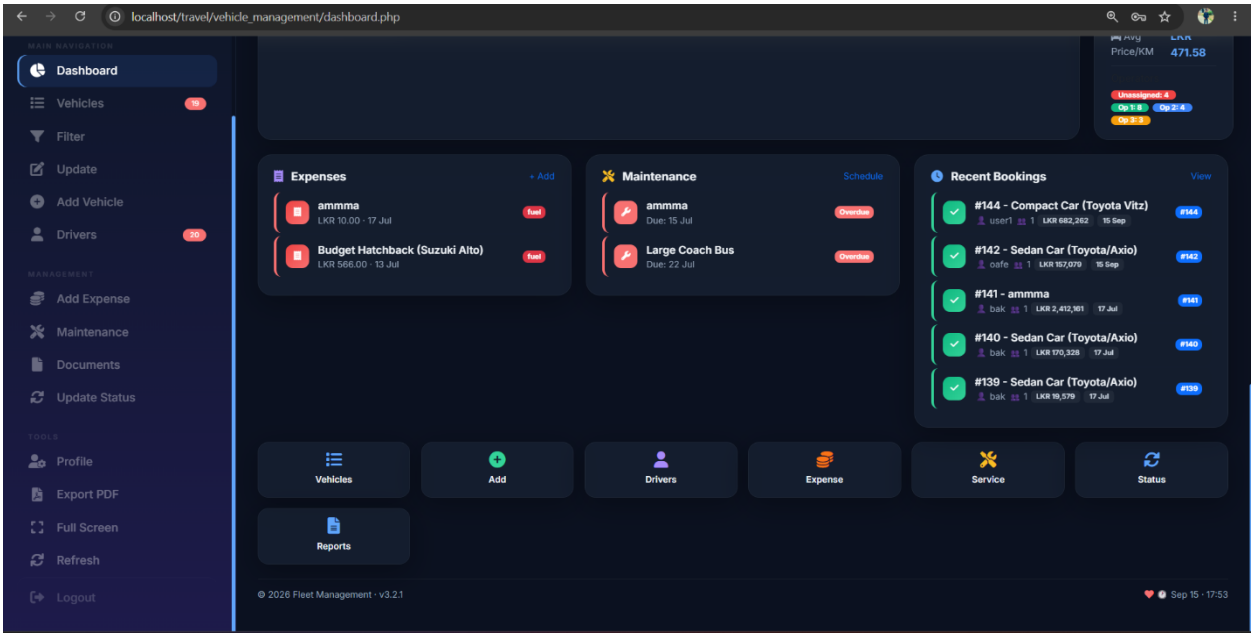
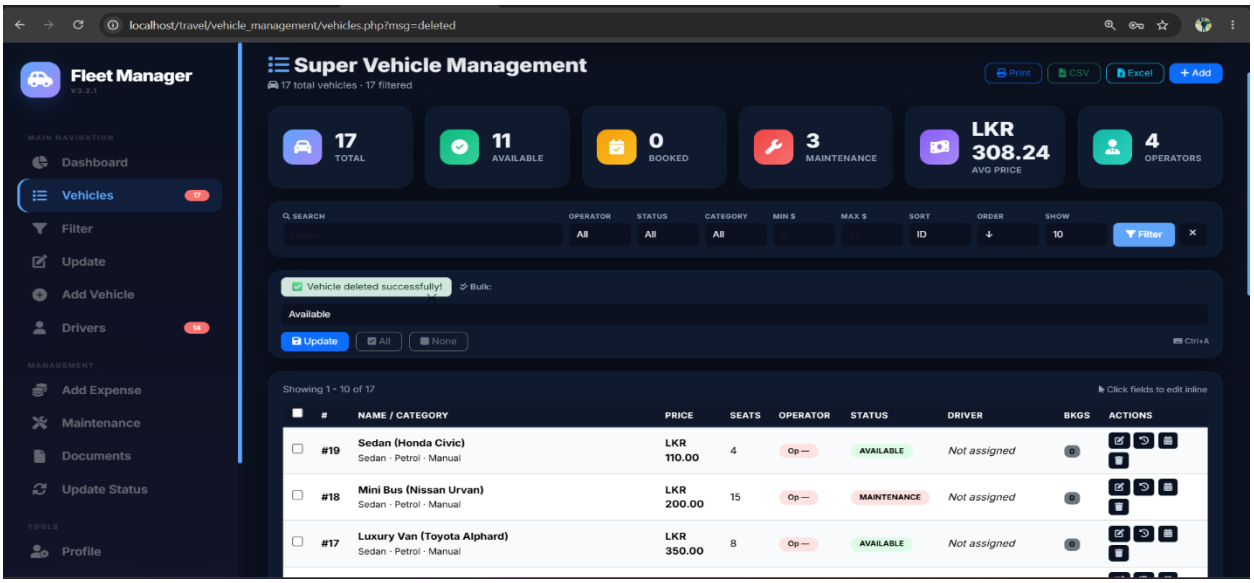
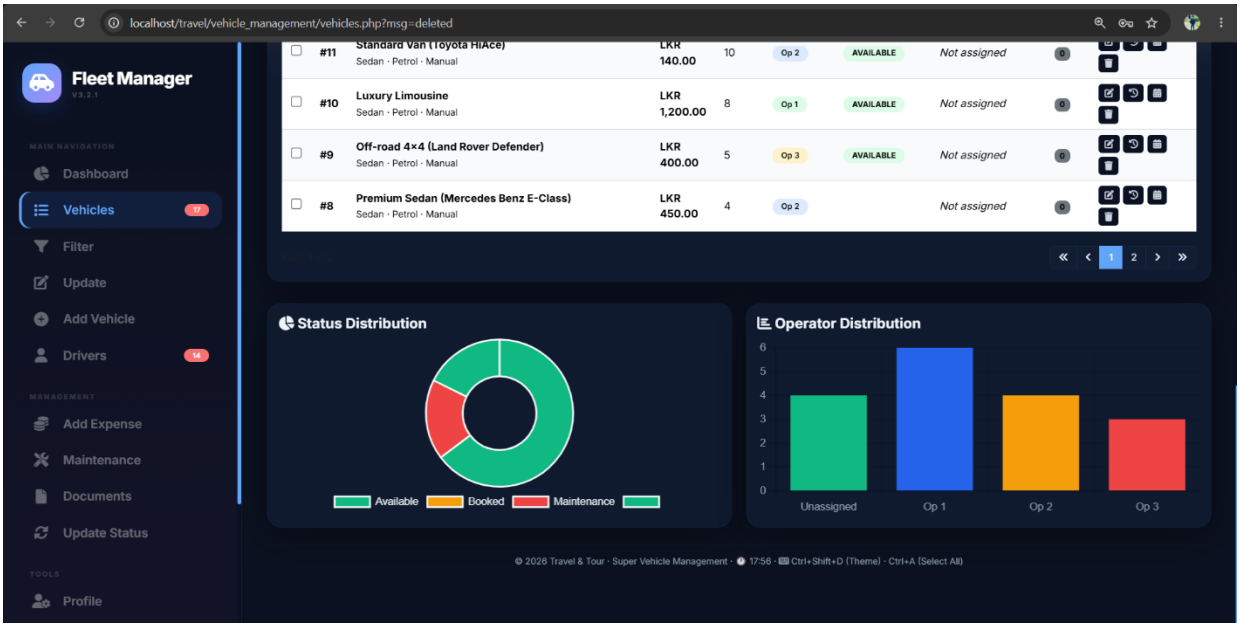
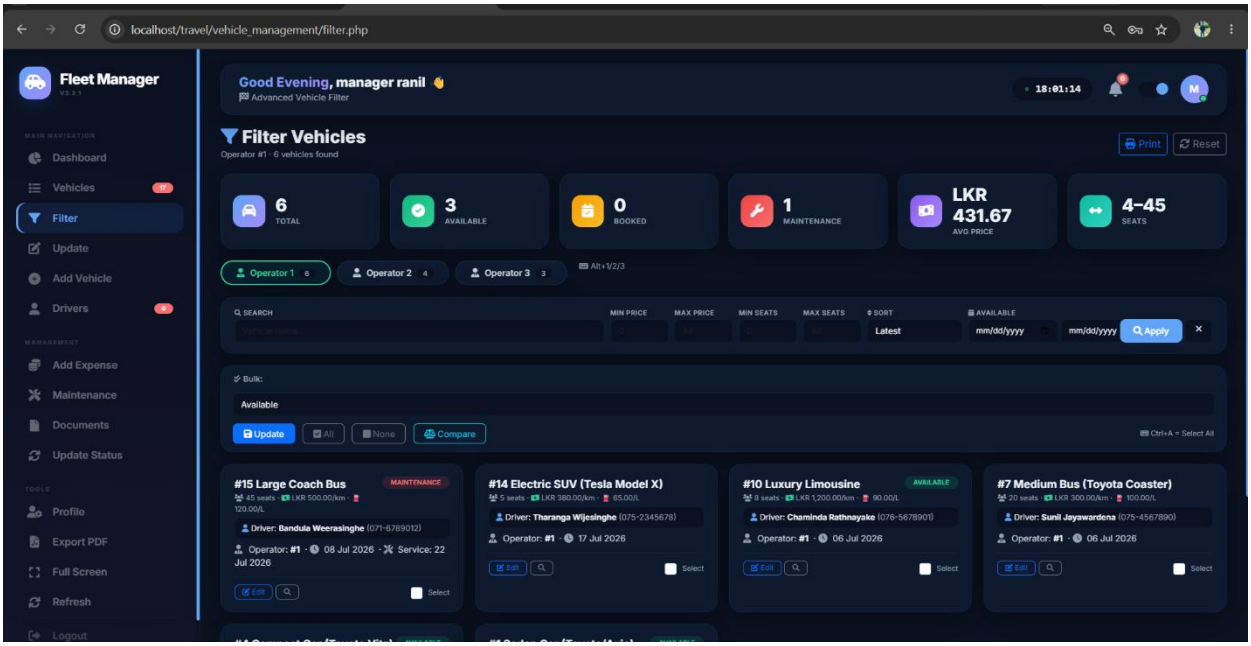


Figure 13 — Vehicle Manager Fleet Management

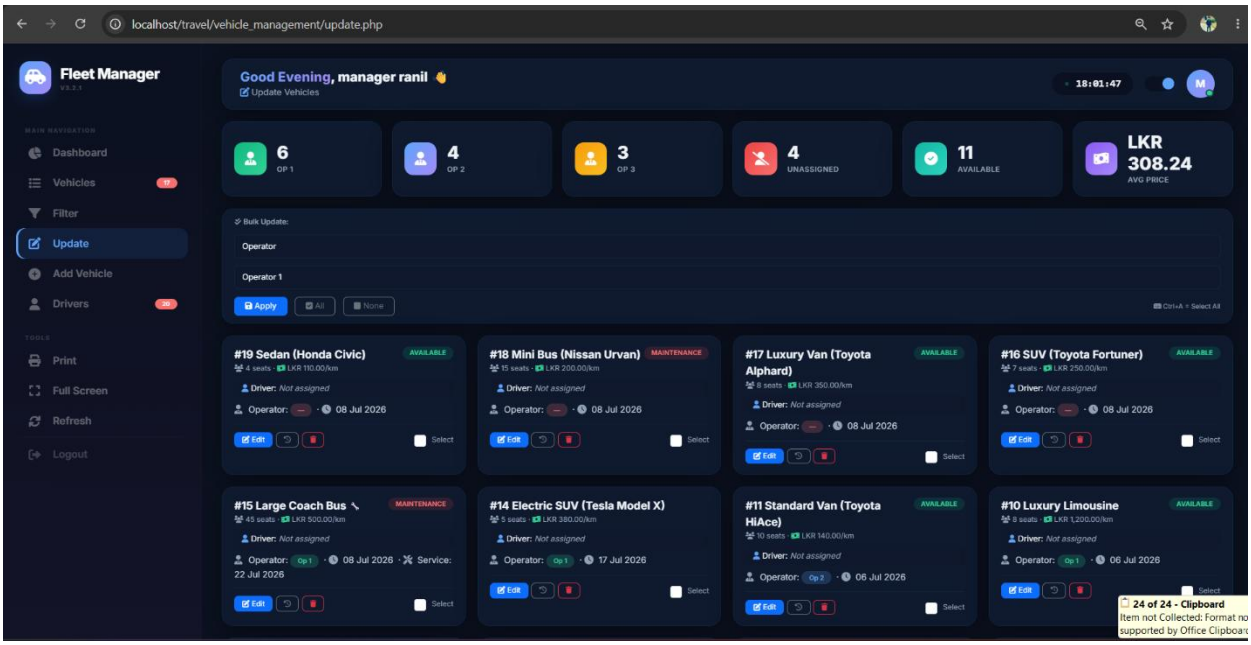




Vehicle filter



Vehicle update



Add vehicle

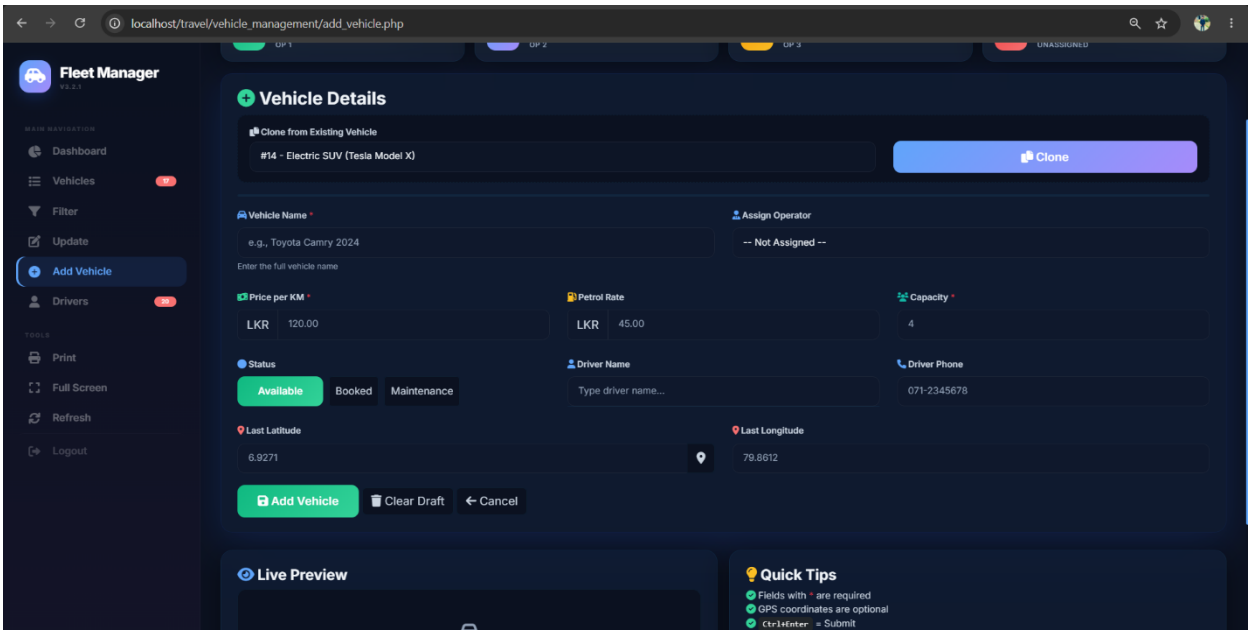
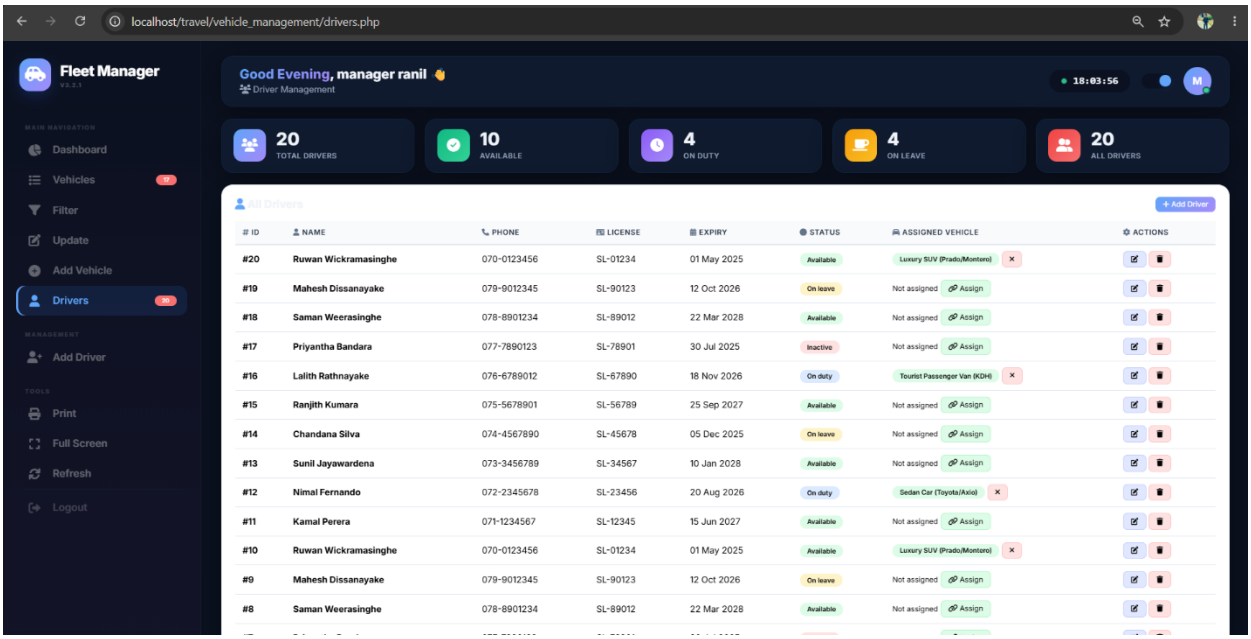
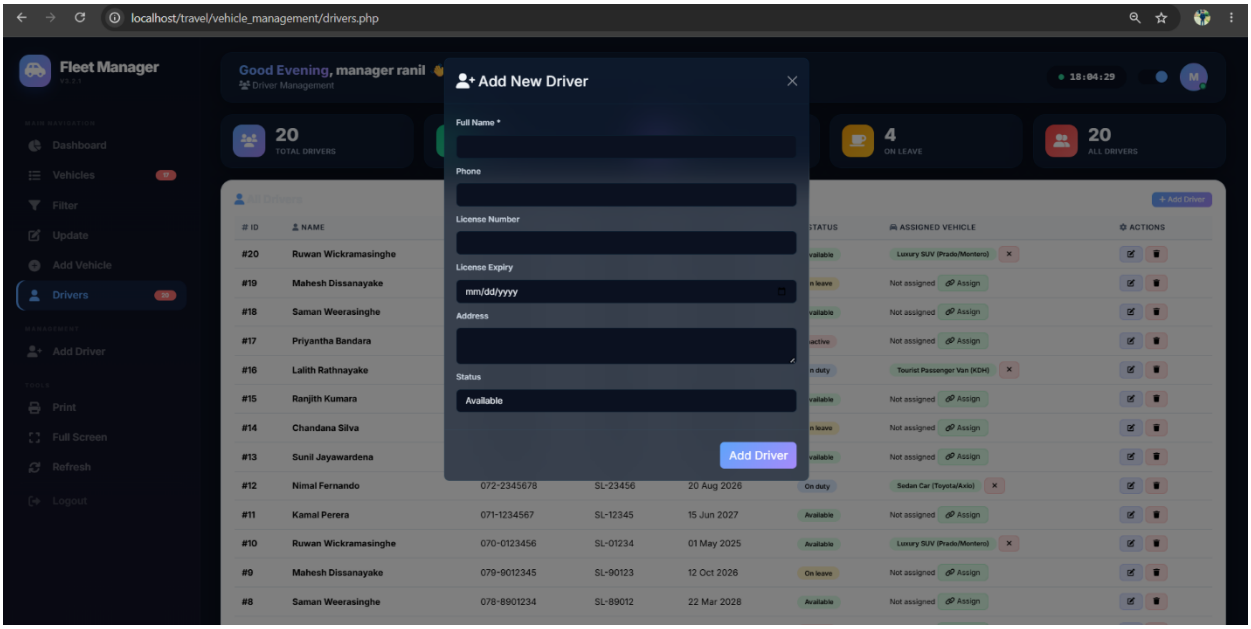


Figure 15 — Vehicle Manager Driver Management



Add new driver



Vehicle driver reports

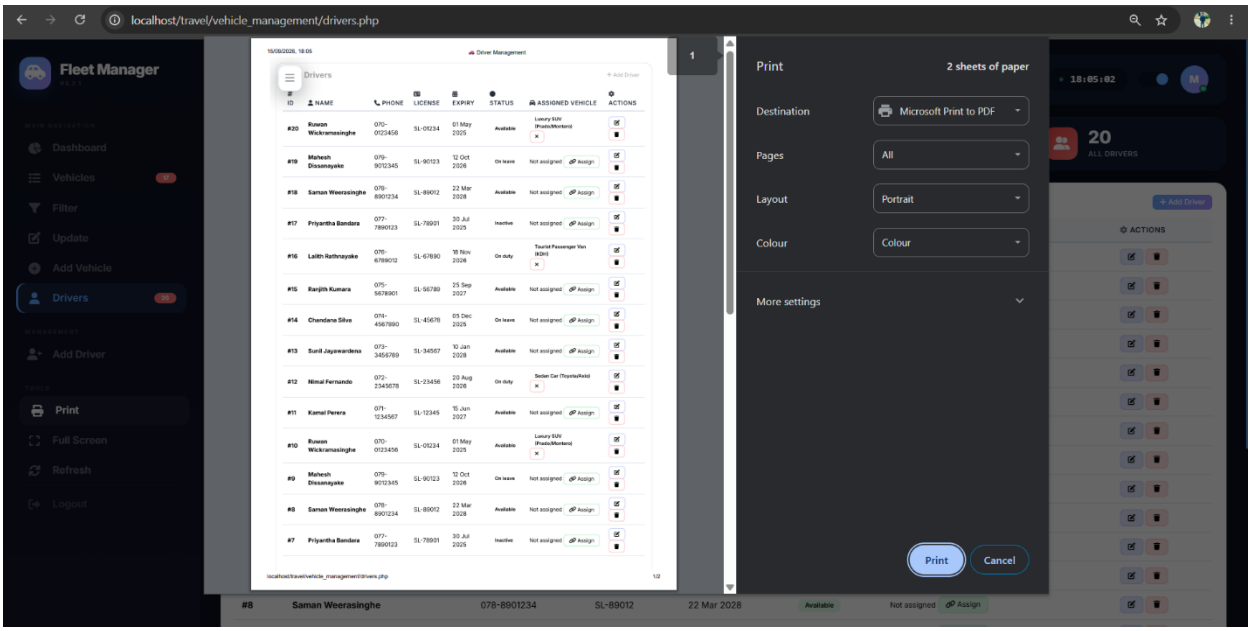
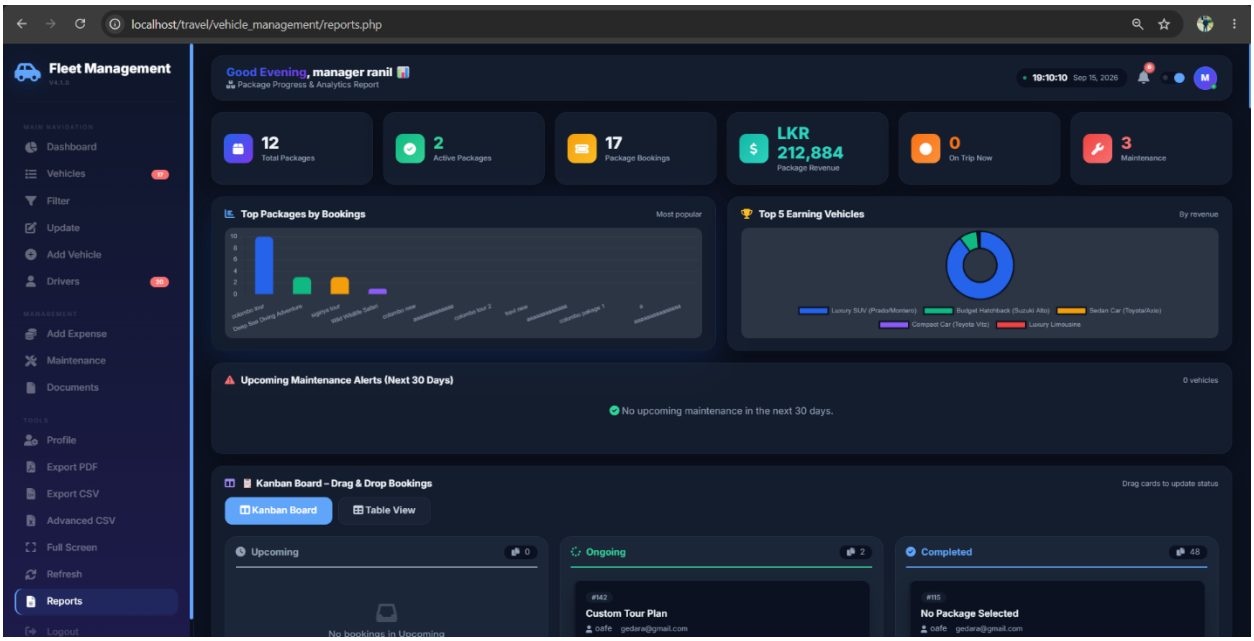
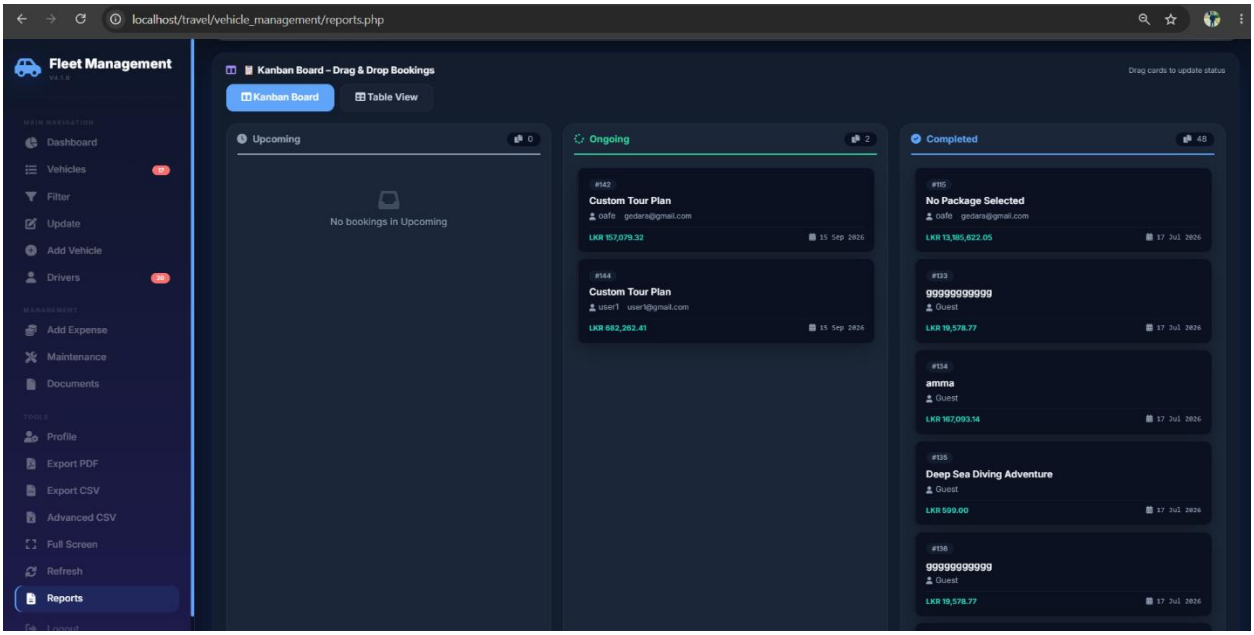
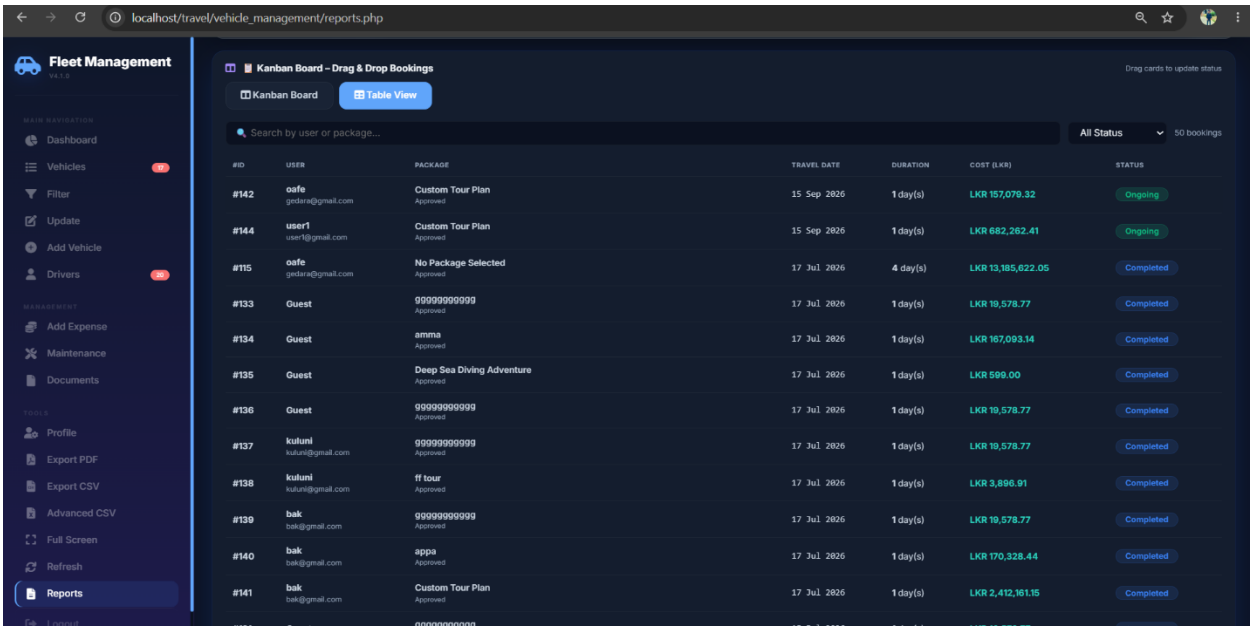


Figure 14 — Vehicle Manager Kanban Reports




localhost/travel/vehicle_management/reports.php
Fleet Management
Kanban Board – Drag & Drop Bookings
Drag cards to update status
Kanban Board
Table View
Upcoming 0
No bookings in Upcoming
Ongoing 2
#142
Custom Tour Plan
oafe gedara@gmail.com
LKR 157,079.32
15 Sep 2026
#144
Custom Tour Plan
user1 user1@gmail.com
LKR 682,262.41
15 Sep 2026
Completed 48
#115
No Package Selected
oafe gedara@gmail.com
LKR 13,185,622.05
17 Jul 2026
#133
ggggggggggg
Guest
LKR 19,578.77
17 Jul 2026
#134
amma
Guest
LKR 167,093.14
17 Jul 2026
#135
Deep Sea Diving Adventure
Guest
LKR 599.00
17 Jul 2026
#136
ggggggggggg
Guest
LKR 19,578.77
17 Jul 2026
Dashboard
Vehicles
Filter
Update
Add Vehicle
Drivers
Add Expense
Maintenance
Documents
Profile
Export PDF
Export CSV
Advanced CSV
Full Screen
Refresh
Reports


localhost/travel/vehicle_management/reports.php
Fleet Management
Kanban Board – Drag & Drop Bookings
Drag cards to update status
Kanban Board
Table View
Search by user or package...
All Status
50 bookings
#ID USER PACKAGE TRAVEL DATE DURATION COST (LKR) STATUS
#142 oafe gedara@gmail.com Custom Tour Plan Aproved 15 Sep 2026 1 day(s) LKR 157,079.32 Ongoing
#144 user1 user1@gmail.com Custom Tour Plan Aproved 15 Sep 2026 1 day(s) LKR 682,262.41 Ongoing
#115 oafe gedara@gmail.com No Package Selected Aproved 17 Jul 2026 4 day(s) LKR 13,185,622.05 Completed
#133 Guest ggggggggggg Aproved 17 Jul 2026 1 day(s) LKR 19,578.77 Completed
#134 Guest amma Aproved 17 Jul 2026 1 day(s) LKR 167,093.14 Completed
#135 Guest Deep Sea Diving Adventure Aproved 17 Jul 2026 1 day(s) LKR 599.00 Completed
#136 Guest ggggggggggg Aproved 17 Jul 2026 1 day(s) LKR 19,578.77 Completed
#137 kuluni kuluni@gmail.com ggggggggggg Aproved 17 Jul 2026 1 day(s) LKR 19,578.77 Completed
#138 kuluni kuluni@gmail.com ff tour Aproved 17 Jul 2026 1 day(s) LKR 3,896.91 Completed
#139 bak bak@gmail.com ggggggggggg Aproved 17 Jul 2026 1 day(s) LKR 19,578.77 Completed
#140 bak bak@gmail.com appa Aproved 17 Jul 2026 1 day(s) LKR 170,328.44 Completed
#141 bak bak@gmail.com Custom Tour Plan Aproved 17 Jul 2026 1 day(s) LKR 2,412,161.15 Completed

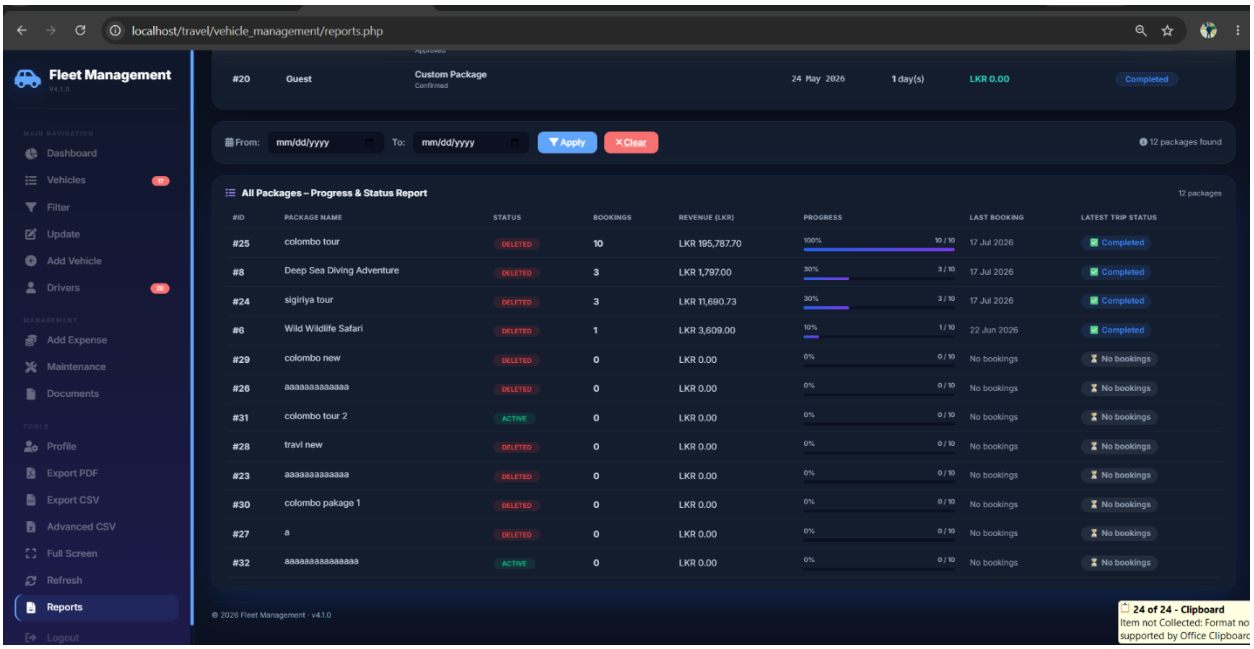


Figure 16 — User Dashboard

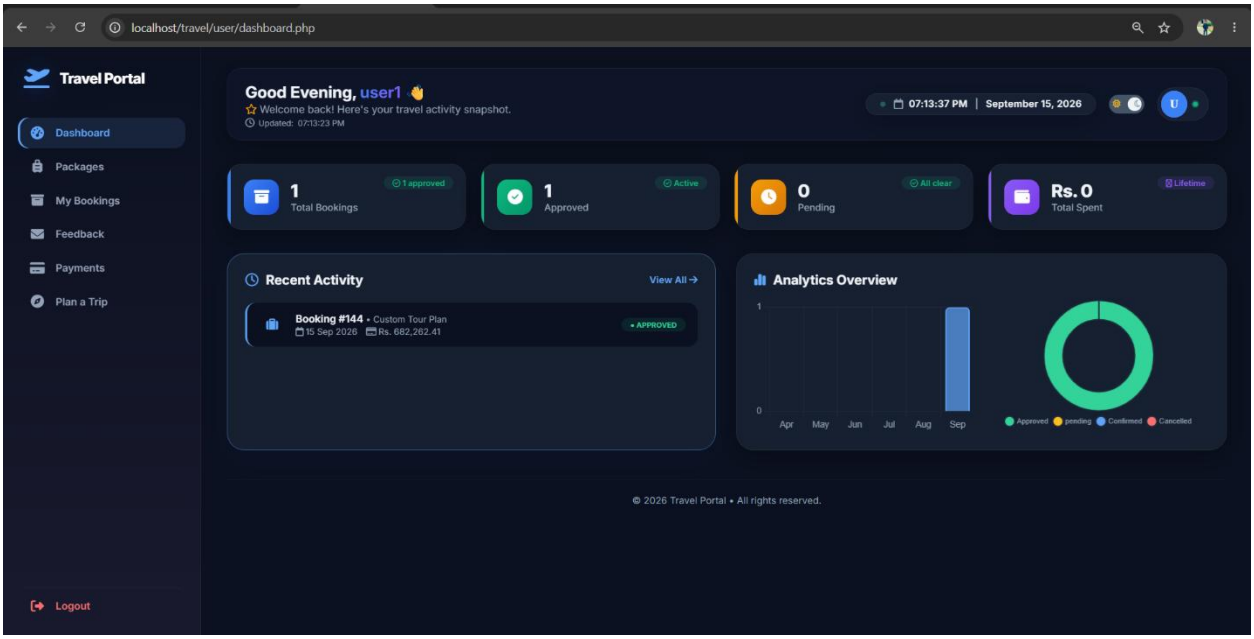


Figure 17 — User Package Browser

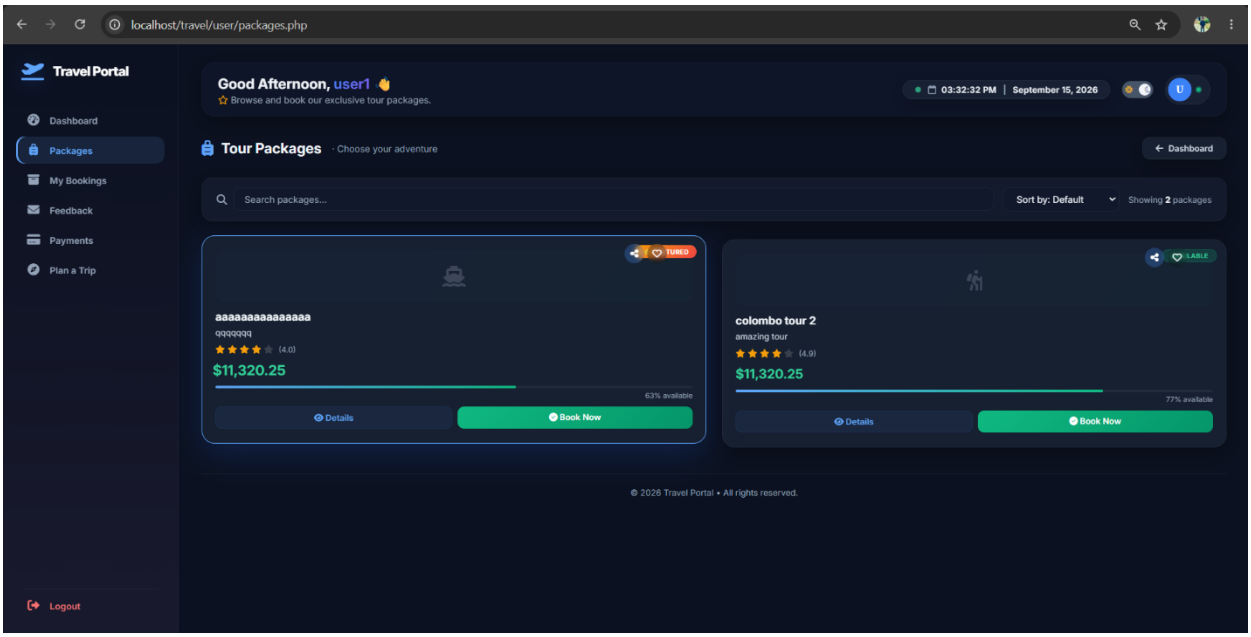


Figure 21 — User Package Details

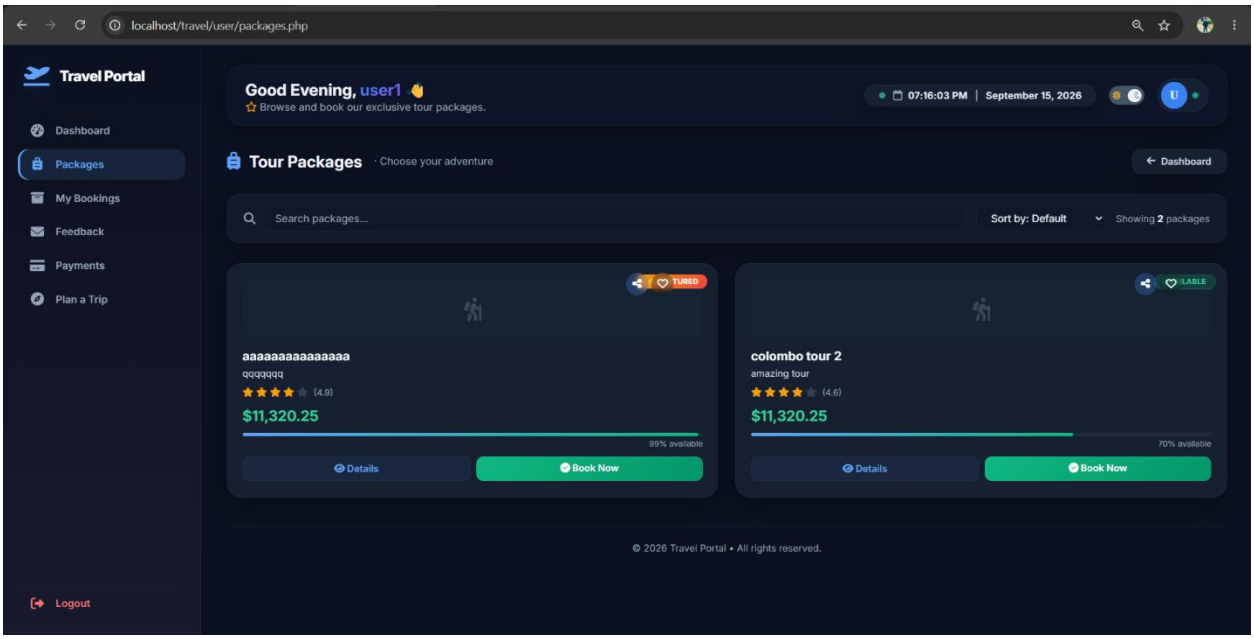
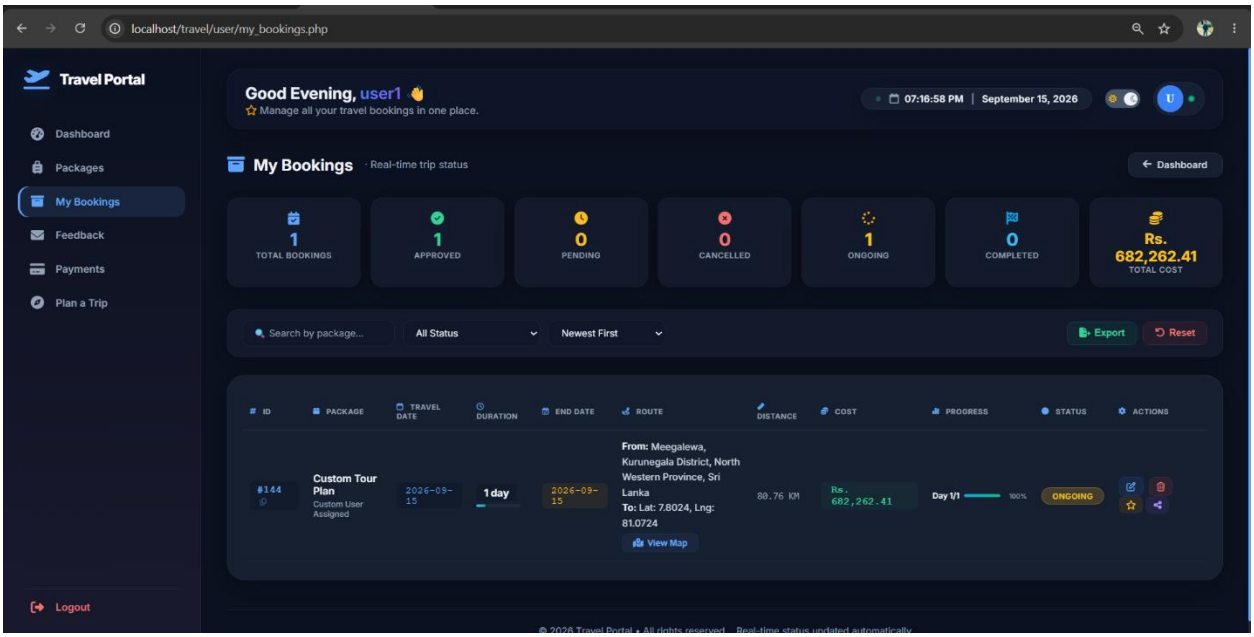
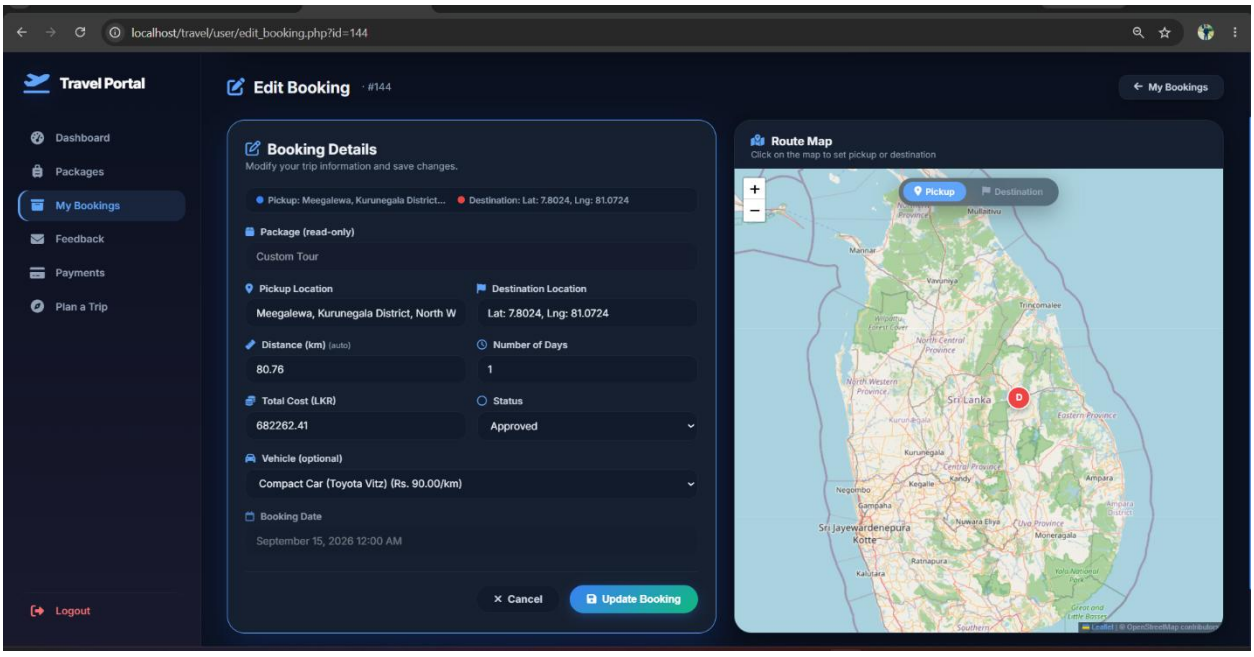


Figure 19 — User My Bookings



Edit bookings user



User feedbacks

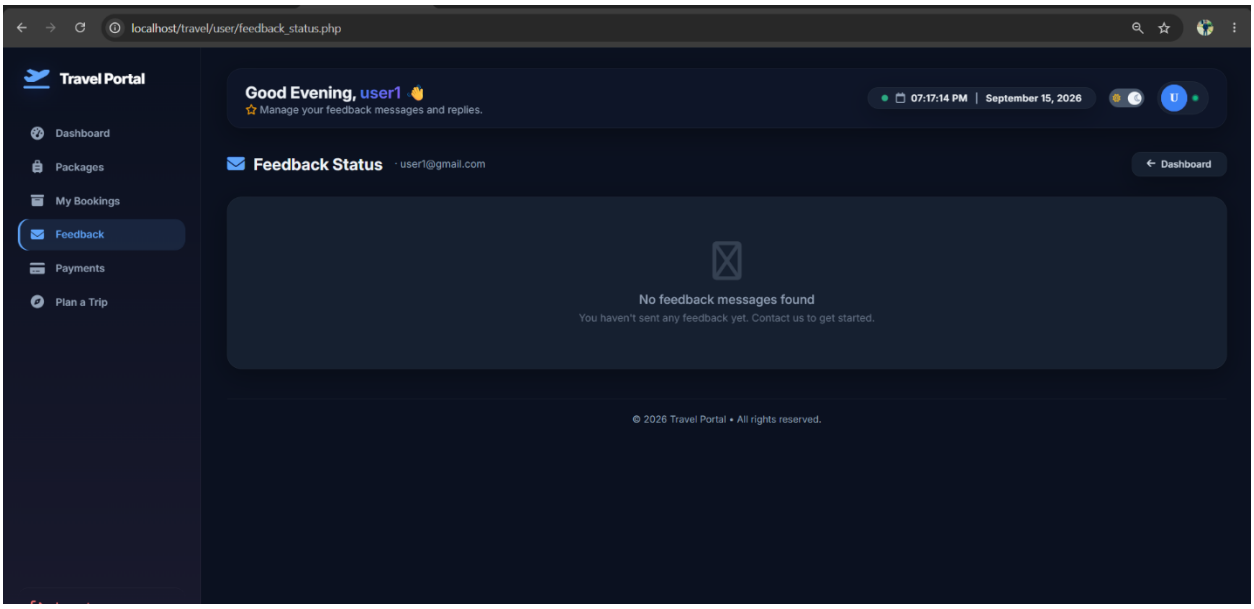


Figure 20 — User Payment Interface

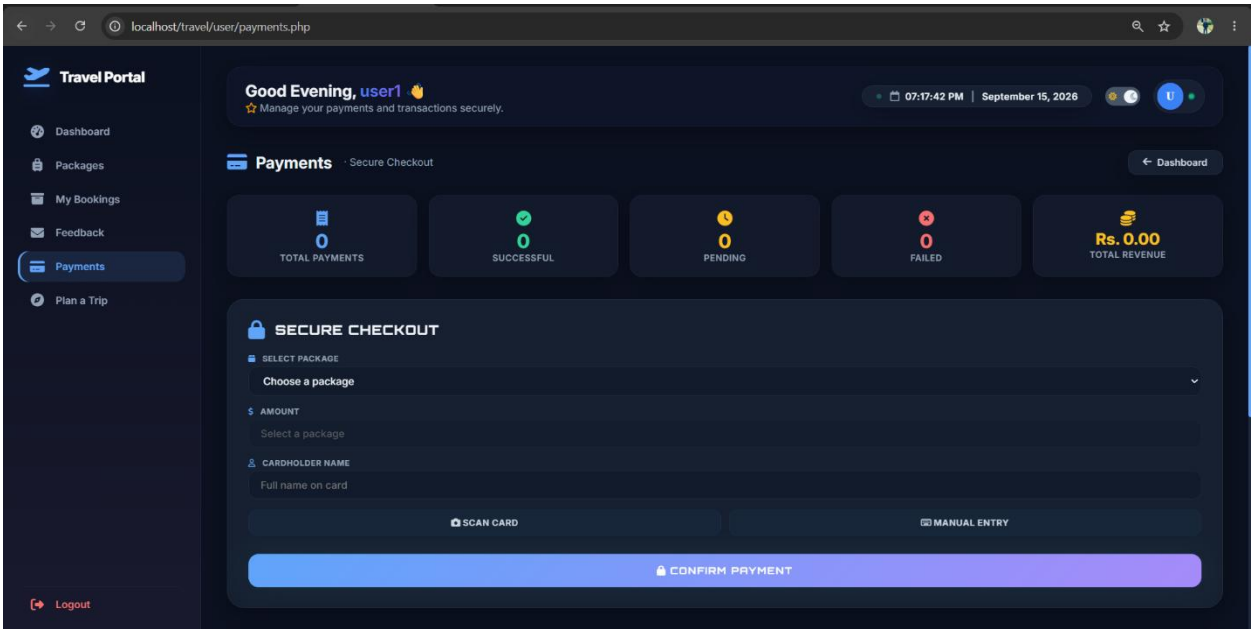
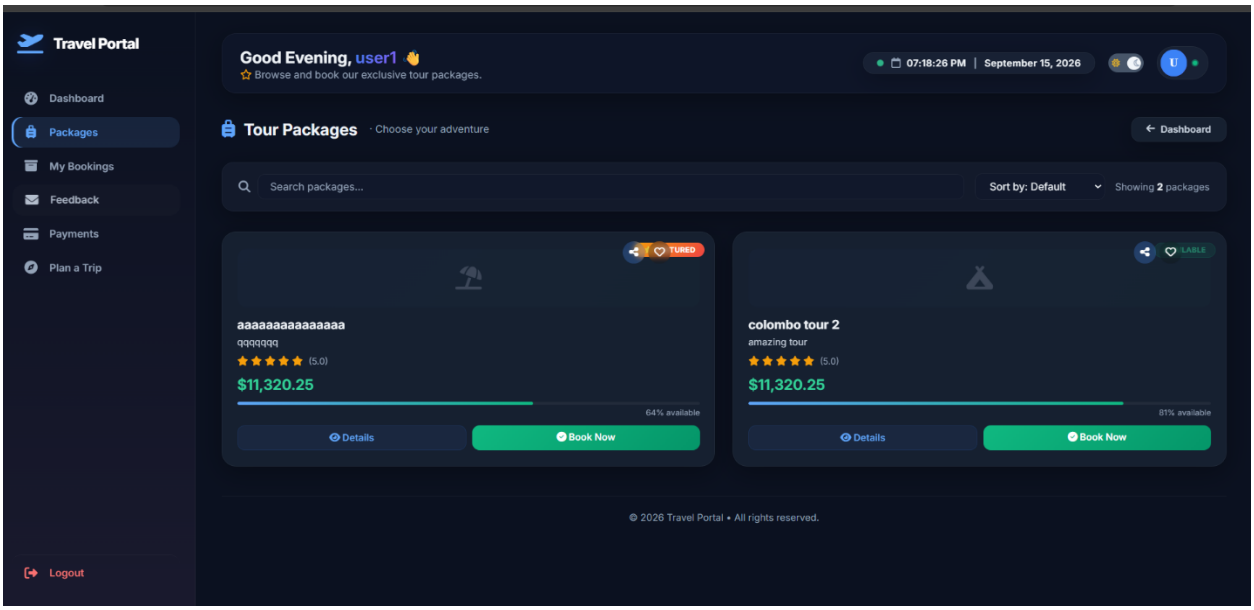


Figure 18 — User Custom Trip Planner



— END OF REPORT —